MATHS in Television

Simon Garner

INSTANT LESSONS

Quality ready-to-use resources

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Simon Garner is a secondary teacher with 15 years of experience, a journalist and educational writer. His teaching background covers a range of schools in Melbourne and Geelong. Simon has taught diverse subject areas to all secondary levels including Mathematics, Science, SOSE, Business Management, VET and Careers.

A working journalist for much of the past 20 years, Simon has written for and edited a number of local suburban, regional and national newspapers and magazines. Over the past five years, he has specialised in educational writing, including scripting more than 200 nationally and internationally distributed educational video/DVD programs, a number of which he also produced.

Simon currently works as Commissioning and Script Editor for VEA, Australia's largest producer of educational DVDs.

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Introduction

Maths in Television is not a textbook. It is a collection of student worksheets that features the application of mathematical knowledge and techniques in a wide range of contexts relating to various aspects of the television industry. These contexts include the types of popular TV programs watched by millions of people nationwide every day and night of the week.

The mathematical areas covered are broadly outlined at the beginning of each worksheet. Many mathematics topics are explored, drawing on a variety of skills and knowledge.

The *Maths In* series has been designed for students in Years 9 and 10. However, the worksheets could be used to motivate and add interest to any Junior or Middle School Mathematics course by demonstrating how mathematics can be applied to everyday life. Note that for most worksheets, students will find a calculator useful.

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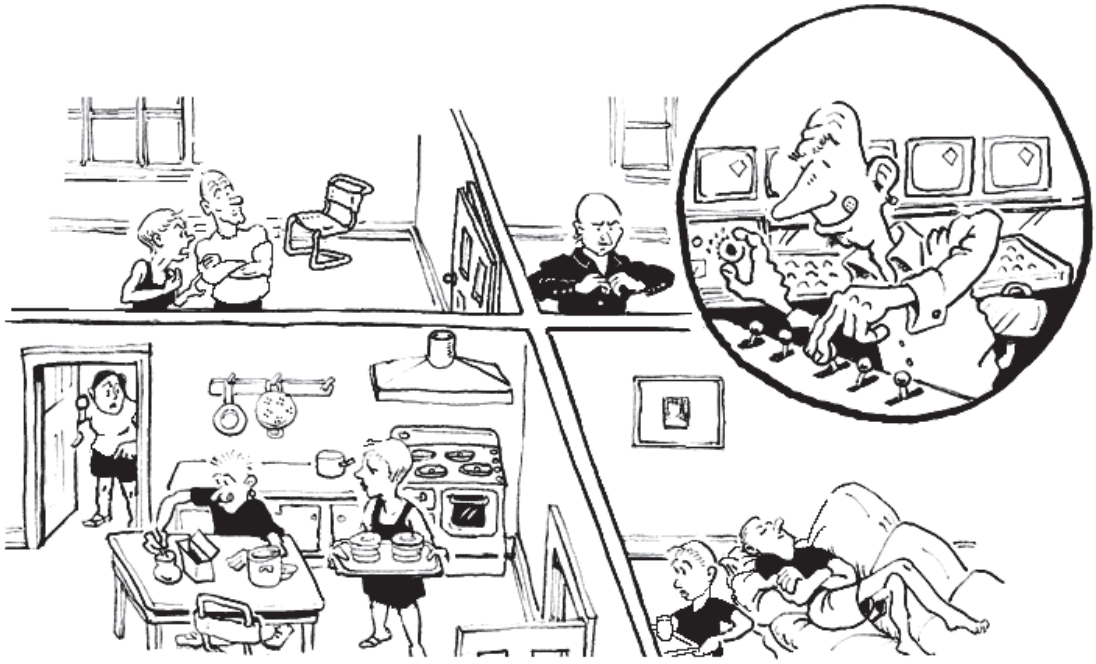
'Little Sister House' 1

1

These housemates are calculating – so there's a bit of problem solving to do, including averages, percentages, ratios and proportions.

A group of housemates is chosen from hundreds of applicants to spend 10 weeks inside the Little Sister House. Viewers watch their trials and tribulations, and have a say in who stays … and who goes.



Unless otherwise indicated, express your answers correct to 2 decimal places.

1 'Little Sister House' starts off with 20 housemates. Their names and ages are:

- Sasha, 22 yrs
- Frida, 19 yrs
- Wendy, 23 yrs
- Isabel, 21 yrs
- Sandra, 21 yrs
- Miffi, 19 yrs
- Rowena, 20 yrs
- Keena, 26 yrs
- Trisha, 22 yrs
- Olivia, 25 yrs
- Jason, 21 yrs
- Reece, 27 yrs
- Steve, 24 yrs
- Tom, 19 yrs
- Marlon, 26 yrs
- Jacques, 23 yrs
- Anton, 24 yrs
- Oscar, 20 yrs
- Kelvin, 25 yrs
- Lenny, 23 yrs

a What is the average (mean) age of the females in the house? Give your answer in years and months, rounded off to the closest whole month.

b What is the average age of the males in the house? Give your answer in years and months, rounded off to the closest whole month.

c What is the average age of all 20 housemates, in years and months?

d What proportion of the housemates is aged 25 and over?

e In simplest form, what is the ratio of teenagers to non-teenagers in the Little Sister House?

f In simplest form, what is the ratio of housemates aged 22 and over to housemates aged 21 and under?

2 At the end of the first week in the house, 20% of the housemates will be evicted. How many of the original housemates will only stay one week in the house?

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3 Each week, every housemate is asked to nominate two people for eviction. The nominations for the first week are listed in the tables below. The six housemates with the most nominations were shortlisted for eviction, with a viewing audience poll used to decide who was evicted. Which six housemates were on the shortlist?

Housemate	Nominees
Sasha	Miffi, Isabel
Frida	Lenny, Sandra
Wendy	Isabel, Marlon
Isabel	Steve, Kelvin
Sandra	Isabel, Marlon
Miffi	Kelvin, Wendy
Rowena	Reece, Isabel
Keena	Miffi, Trisha
Trisha	Anton, Sandra
Olivia	Reece, Marlon

Housemate	Nominees
Jason	Trisha, Isabel
Reece	Marlon, Anton
Steve	Oscar, Miffi
Tom	Reece, Olivia
Marlon	Isabel, Anton
Jacques	Marlon, Miffi
Anton	Reece, Sandra
Oscar	Isabel, Wendy
Kelvin	Oscar, Marlon
Lenny	Reece, Miffi

4 At the end of the second week, another two housemates were evicted. There were four nominations for eviction: Reece, Kelvin, Wendy and Sandra.

a Two of these housemates were nominees at the end of the first week. Who are they?

b Name the four housemates evicted at the end of the first week.

c At the end of the second week, the votes from the viewing audience for three of the nominees were:

Reece ..... 28% of votes

Kelvin .... 18% of votes

Wendy ... 34% of votes

What percentage of the votes did Sandra receive?

d Which two housemates were evicted at the end of the second week?

e A total of 156 750 votes was received from the viewing audience at the end of the second week. Calculate the number of votes each of the four nominees received.

f What was the difference between Reece's votes and Kelvin's votes?

5 At the end of the third week, another two housemates were evicted: Oscar and Lenny.

a What proportion of the original housemates were now left in the house?

b In simplest form, what was the ratio of males to females in the house at the end of the third week?

6 During the fourth week, the number of housemates increased to 16. How many new housemates were introduced during this week?

7 After the new housemates were introduced during the fourth week, the ratio of males to females in the house was 3 : 5. How many of the new housemates were males? How many were females?

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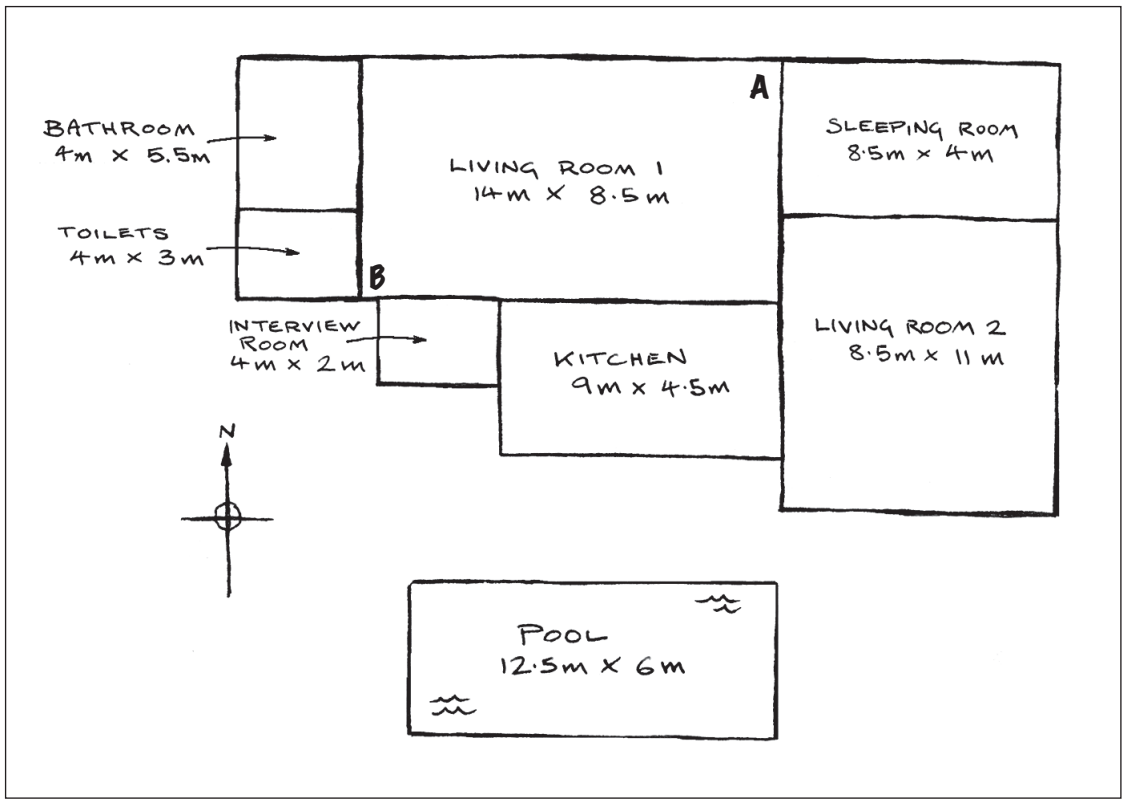
'Little Sister House' 2

A dream house on a tropical island has a range of problems involving area, volume, speed, time, directions and Pythagoras' theorem.

The 'Little Sister House' is built on a tropical island just off the northern Australian coast. Cameras pick up the action as they adjust to sharing the house and taking part in 'Little Sister's' game, which has many twists and turns.

Here is a floor plan of the 'Little Sister House'.

All room measurements shown are east–west length × north–south length.



Map not to scale

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'Little Sister House' 2 (cont'd)

2

Unless otherwise indicated, express answers correct to 2 decimal places.

1 **a** Calculate the total area of the living rooms in the Little Sister House.

b If there are 20 housemates in Living Room 1, how much floor space does each one have?

2 Which has the greater floor area: the Sleeping Room or the Bathroom and Toilets combined?

3 One door to the Sleeping Room is in the corner marked A on the floor plan. The door into the Toilets is in the diagonally opposite corner, marked B on the floor plan.

How far is it in a straight line from the Sleeping Room to the Toilets (A → B)?

4 What is the total area of the Little Sister House?

5 There is one camera placed in the house for every 3.5 m^2 of floor space. No cameras are placed in the toilets.

Calculate the number of cameras in the house (rounding off to the nearest whole camera).

6 One housemate swam 12 laps of the length of the pool. How far did she swim in total?

7 Another housemate swam 7 laps in 1 min 52 s.

a What was his average time per lap (in seconds)?

b What was the total distance he swam?

c What was his average speed for the 7 laps (in m/s)?

8 **a** The pool has an average depth of 1.7 metres. What volume of water does it hold? (Answer in m^3.)

b How many litres is this?

9 What direction

a is the Sleeping Room from the Interview Room?

b are the Toilets from the Bathroom?

c is the Kitchen from Living Room 2?

d is Living Room 1 from the Pool?

e is the Bathroom from Living Room 2?

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Game shows

3

Cash and prizes! Come on down for problem solving, logical thinking, probability and using algebraic formulae.

Lots of glitter and hype, a wheeling and dealing host, and an excited studio audience – there's lots of cash and prizes up for offer. Who'll strike it lucky on tonight's TV game shows?

Unless otherwise indicated, express your answers correct to 2 decimal places.

1 In the TV game show 'It's Tempting', contestants answer general knowledge questions by being first on the buzzer in three separate rounds. They receive $10 for each correct answer in round one; $15 for each correct answer in round two; and $20 for each correct answer in round three. In any round, an incorrect answer costs them $10. Each contestant also has the opportunity to spend money to purchase products.

Here are the results of one night's contest:

Contestant	Round 1 No. of correct answers	Round 2 No. of correct answers	Round 3 No. of correct answers	Total no. of incorrect answers	Money spent
Jasmine	3	5	2	3	$40
Huan	4	1	2	2	$25
Adam	2	1	4	1	$35

Use the results above to complete the table below. Then calculate the total number of dollars each contestant received to work out who came first, second and third this night.

Contestant	Round 1 $ won	Round 2 $ won	Round 3 $ won	Minus $ for incorrect answers	Minus money spent	Total $ for the night	Rank (1st, 2nd or 3rd)
Jasmine					$40		
Huan					$25		
Adam					$35		

2 A TV game show called 'The Right Price' involves contestants ranking products in order from lowest price to highest price. Here is a list of the products from one show and, in random order, their monetary values.

Products:

- Car
- Mountain bike
- Home entertainment system
- Lounge room furniture
- Holiday
- Gym equipment
- Luggage

Monetary values:

$35 000, $2 500, $4 500, $14 000, $17 500, $9 000, $1 750

Use the following information to work out the monetary value of each product, then write it beside that product.

- The lounge room furniture was worth exactly half as much as the home entertainment system.
- Nothing was more expensive than the car.
- The luggage was more expensive than just one other item, but not as expensive as the gym equipment.
- The holiday was worth exactly eight times as much as one of the other items.

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3 In the game show 'Let's Make a Deal', contestants are given a choice of 24 mystery boxes. They select one and put it aside, then eliminate the other 23 during the show.

Six mystery boxes contain money prizes worth \$10 000, \$20 000, \$25 000, \$50 000, \$75 000 and \$100 000. Four mystery boxes contain holidays; four contain home appliances; two contain cars; and the rest contain dud prizes such as paper clips, chewing gum and ping-pong balls.

At four different stages during the show, contestants are offered a deal from the host – take an amount of money and leave, or keep playing and receive the contents of the box they originally selected.

Use this information to answer the following questions. Express all your answers in simplest form.

Janice has selected a box to put aside and no other boxes have been eliminated.

a What is the probability that she selected a money prize?

b What is the probability that she selected a money prize greater than \$40 000?

c What is the probability that she selected a car?

d What is the probability that she selected a holiday or a home appliance?

e Which is the greater probability: Janice selected a dud prize or Janice selected a money prize?

f How many times greater is the probability that Janice selected a home appliance than the probability that she selected a money prize worth less than \$15 000?

4 When Darryl the host offers deals to the contestants, the size of the deal is based on the total prize value of the remaining mystery boxes. This total prize value is known to Darryl but not to the contestants. The deal figure is calculated according to this secret formula:

Deal = Total prize value of remaining boxes multiplied by 0.4 divided by 2

$$\text{or } D = \frac{V \times 0.4}{2}$$

Where: D = Deal, V = Total prize value of remaining boxes

a If the total prize value in the remaining boxes is \$36 000, what is the value of the deal?

b If the \$50 000 and \$20 000 boxes are eliminated, and the remaining non-cash prizes have a total value of \$40 000, what is the value of the deal?

c At one stage in the game, Darryl offers a deal of \$37 000. What is the total value of the prizes in the remaining boxes?

d What is the total value of remaining prizes if Darryl offers a deal of \$15 000?

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'The Best Loser' 1

4

Shedding the kilos in front of a national audience – the ups and downs of percentages, rates of change, unit conversion, using the BMI (Body Mass Index formula) and graphing.

Five overweight contestants – they all want to change their habits and lose their excess fat … They're doing it on TV so people at home can sit on the couch and watch their progress, and there's more than a bit of maths involved.

Here are the contestants and their details at the initial weigh-in:

Contestant	Height (cm)	Weight (kg)
Susan	153	85
Heidi	167	91
Stefan	179	128
Mohammed	181	115
Frank	194	133

While there is no one recommended weight-for-height figure, scientists recommend a range of weights for height. These are shown for each contestant in the table to the right.

Height (cm)	Weight range (kg)
153	47–58
167	56–70
179	64–80
181	66–82
194	75–94

Unless otherwise indicated, express answers correct to 1 decimal place.

1 Why do you think that ranges of weights are used in the table above, and not one specific weight figure?

2 After careful assessment by a doctor, each contestant is given a target weight to reach after six months of dieting and exercise. These are shown in the table at right.

Contestant	Target weight
Susan	53 kg
Heidi	68 kg
Stefan	83 kg
Mohammed	81 kg
Frank	91 kg

Draw a bar graph to show each contestant's initial weight and target weight.

3 Rank the contestants in order from highest to lowest in terms of the number of kilograms they must lose to achieve their target weight.

4 **a** Calculate the percentage of their current body weight that each contestant has to lose to achieve their target weight. Use this equation:

$$\%\ \text{weight loss} = \frac{\text{initial weight} - \text{target weight}}{\text{initial weight}} \times 100$$

Contestant	% weight loss
Susan	
Heidi	
Stefan	
Mohammed	
Frank	

b Rank the contestants in order from highest to lowest in terms of these percentage weight loss figures.

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5 If each contestant takes exactly six months to reach their target weight, calculate the average rate of weight loss in kg/month.

Contestant	Average rate of weight loss (kg/month)
Susan	
Heidi	
Stefan	
Mohammed	
Frank	

6 Medical experts use a measure known as Body Mass Index, or BMI, as a more accurate way of measuring the relationship between a person's weight and their height. It is calculated according to the following formula:

$$\text{BMI} = \frac{\text{weight in kg}}{(\text{height in metres})^{\text{squared}}} \quad \text{or:} \quad \text{BMI} = \frac{\text{kg}}{\text{m}^2}$$

a How do you convert height from centimetres to metres?

b Calculate each contestant's BMI at the initial weigh-in.

c Calculate each contestant's BMI once they reach their target weight.

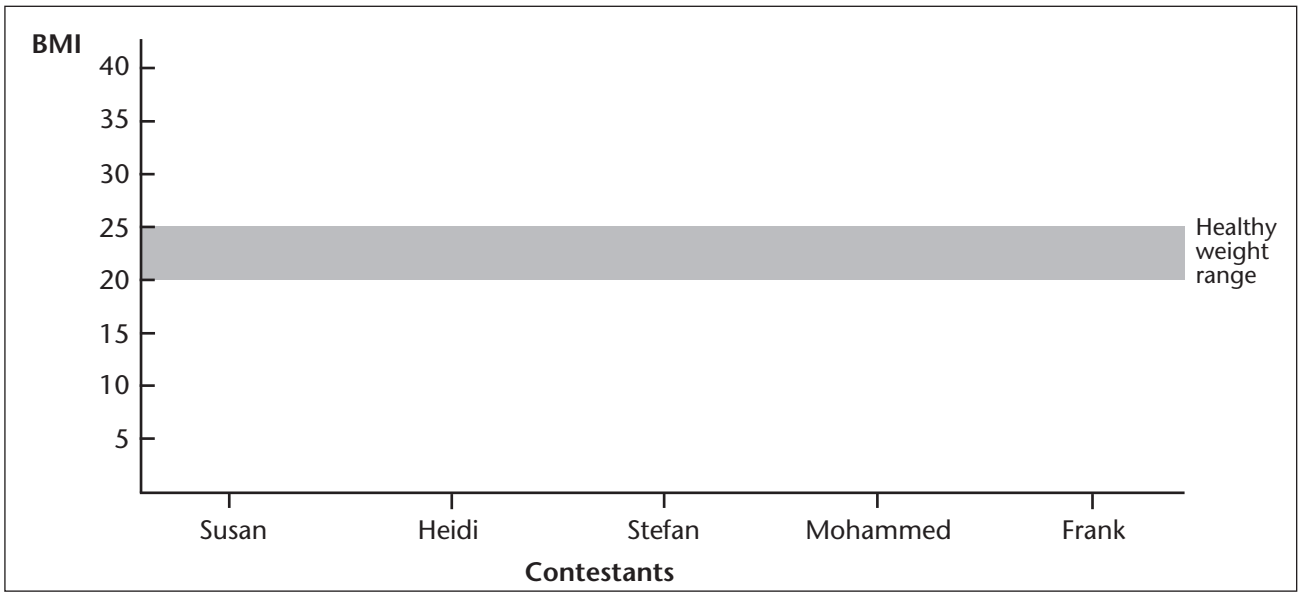
7 Body mass index categories:

- A person with a BMI between 20 and 25 is considered to be of **healthy weight**.
- A person with a BMI between 26 and 30 is considered to be **overweight**.
- A person with a BMI greater than 31 is considered to be **obese**.
- A person with a BMI of less than 19 is considered to be **underweight**.

a Into which BMI category (healthy weight, overweight, obese, underweight) did each contestant fit at the initial weigh-in?

b If all contestants reached their target weight, would this place them in the healthy weight category?

8 On this graph, indicate with accurately placed points the BMI for each contestant at initial weigh-in, and the BMI when their target weight is reached. Use one colour for the initial BMI and a different colour for the target weight BMI. Include a title for the graph.



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'The Best Loser' 2

Who loses weight the quickest ... and who puts it on? Our contestants are sweating it out and burning up the kilojoules. But do they burn up enough kilojoules to shed the kilograms?

The contestants participating in 'The Best Loser', together with their initial and target weights, are given below.

Contestant	Initial weight (kg)	Target weight (kg)
Susan	85	53
Heidi	91	68
Stefan	128	83
Mohammed	115	81
Frank	133	91

The contestants' weights after following a diet and exercise program for one month are shown in the table to the right.

Contestant	Weight (kg)
Susan	80.5
Heidi	87
Stefan	116
Mohammed	110
Frank	117

1. How much weight did each contestant lose over the month?
2. Rank the contestants in order from highest to lowest according to the number of kilograms lost.

To lose fat, the body must expend or 'burn' more energy than it takes in. Expenditure of energy occurs during exercise.

Energy is taken into the body through eating food. Energy expended, and taken in, is measured in kilojoules (kJ).

People of different weights burn kilojoules at different rates. This table shows approximations of how many kilojoules are burned during different activities.

ACTIVITY	kJ burned per hour, per kg of body weight (approx)
Sitting	4
Standing	6
Housework	14
Slow walking	17
Golf	19.5
Brisk walking	20
Basketball	26
Tennis	26
Swimming	30
Cycling	35
Jogging	37
Circuit training	42.5

After consultation with a dietician and exercise physiologist, the five contestants on 'The Best Loser' have a recommended daily energy intake, together with an exercise program. These are shown in the table below.

Contestant	Daily energy intake
Susan	8 750 kJ
Heidi	9 250 kJ
Stefan	10 750 kJ
Mohammed	11 250 kJ
Frank	12 500 kJ

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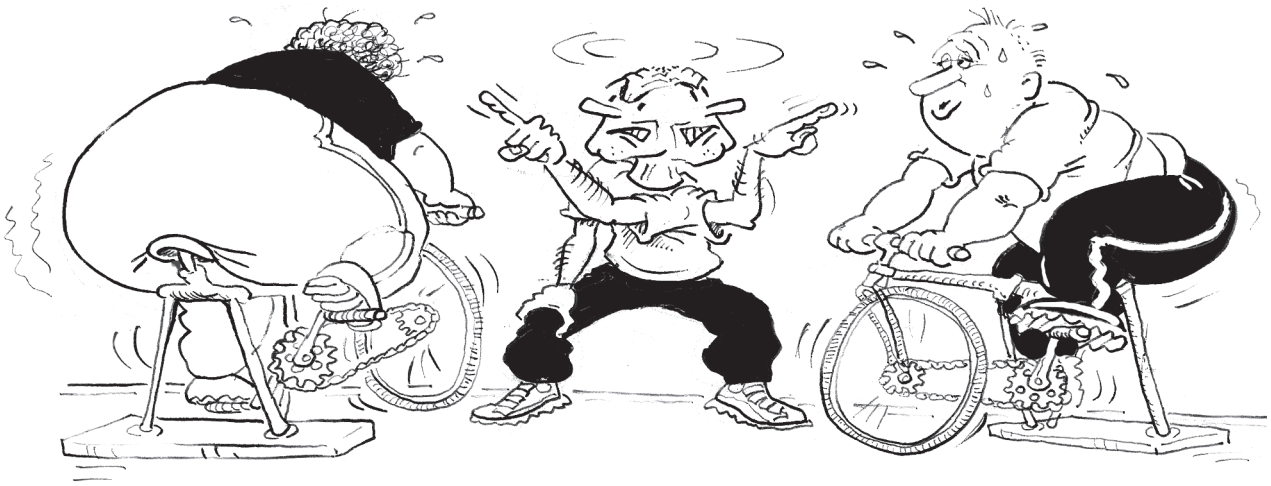

3 Heidi, Stefan and Frank all consumed their recommended daily energy intake in one day and were involved in the activities shown. Calculate which of the three would effectively lose weight that day by working out the total energy expended compared with the energy intake. If expended energy is greater than energy intake, weight is effectively lost.

Heidi	
Activity	**Energy expended**
1 hour of cycling	
4.5 hours of sitting	
1.5 hour of slow walking	
4 hours of standing	

Stefan	
Activity	**Energy expended**
4 hours of golf	
1 hour of tennis	
5 hours of sitting	
3.5 hours of standing	

Frank	
Activity	**Energy expended**
0.5 hours of jogging	
0.25 hours of brisk walking	
6.5 hours of sitting	
2 hours of standing	

Hint:

Remember the energy expended figures in the table on the previous page are kJ burned per kg of body weight.

In your calculations, use the contestants' initial weights as given at the beginning of this activity sheet.

4 By how much did each contestant's expended energy exceed, or fall short of, their energy intake?

5 Construct a line graph on graph paper to show the different amounts of energy taken in and expended by each of these contestants. Put both lines on the same graph, but use a different colour for each one. Make sure you label both lines and the axes clearly, and give the graph a title.

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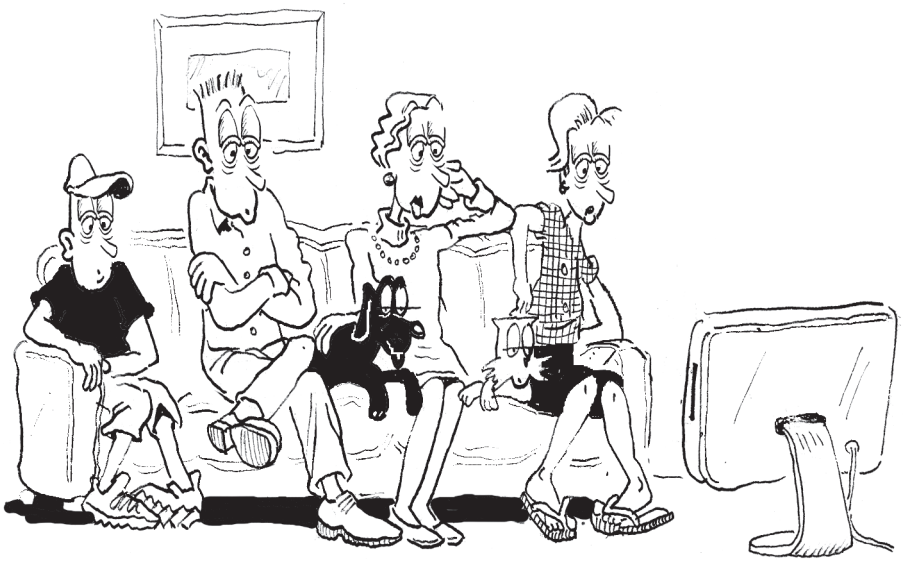
Advertising

6

Proudly brought to you by a range of calculations involving advertising times, rates, revenue and marketing methods.

Billions of dollars are spent every year on television advertising. This money allows commercial free-to-air television stations to exist. Many viewers find advertisements irritating, but television advertising is a highly effective way for companies, corporations, governments and non-profit organisations to get their messages across to the public.



Unless otherwise indicated, express answers correct to 2 decimal places.

1 When television advertising is purchased, it is sold in packages, not as single advertisements. Different advertising rates are charged according to the city or region in which they screen, and the time of day they are shown.

a Why do you think advertising rates vary according to the city or region in which they are shown?

b Why do you think advertising rates vary according to the time of day they are shown?

2 The basic time unit for advertisements is 30 seconds. Channel 4 charges the following rates for advertisements shown between 7 am and 9 am:

- In Melbourne – $19 000 for 30 × 30-second advertisements
- In Sydney – $21 500 for 30 × 30-second advertisements
- In Brisbane – $14 200 for 30 × 30-second advertisements
- In Adelaide – $5 200 for 30 × 30-second advertisements
- In Perth – $6 700 for 30 × 30-second advertisements

Use these rates to calculate how much advertisers pay per advertisement on Channel 4 in each city.

3 An advertiser purchases 30 × 30-second advertising spots to run between 7 am and 9 am over the course of two weeks (i.e. Monday to Friday). On average, how many times per day would the advertisement be shown?

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4 While 30 seconds is the basic time unit for an advertisement, advertisers can purchase 15-, 45- and 60-second spots too.

- 15-second advertising spots are charged at 60% of the 30-second rate
- 45-second advertising spots are charged at 140% of the 30-second rate
- 60-second advertising spots are charged at 160% of the 30-second rate

Fill in the table below, given that the advertisements are airing on Channel 4 in Brisbane between 7 am and 9 am.

Advertisement length	Cost of 30 ads	Cost of 1 ad
15 seconds		
45 seconds		
60 seconds		

5 Prime weekday viewing times are from 6 pm to 9.30 pm. During these times, a 50% loading is added to the 7 am to 9 am costs. Between midnight and 7 am, the advertising costs are 45% of the 7 am to 9 am costs.

An advertiser wants to run this advertising schedule on Channel 4 in Perth:

- 30 × 30-second advertisements between 7 am and 9 am
- 30 × 15-second advertisements between 6 pm and 9.30 pm
- 30 × 45-second advertisements between midnight and 7 am

What would be the total cost of the campaign?

6 An advertiser ran 30 × 30-second advertisements on Channel 4 in Adelaide between 7 am and 9 am across two weeks. The advertisements offered a special deal on a product, which lasted for the length of the advertising campaign.

The deal let viewers purchase the product at 50% of its regular price. The regular price was $59. The advertising campaign resulted in 216 sales of the product.

a Was the revenue gained from product sales during the advertising campaign more or less than the cost of the campaign?

b By how much did the revenue gained from the product sales exceed, or fall short of, the cost of the campaign?

7 A movie is scheduled to screen between 8.30 pm and 10.30 pm on a commercial TV station. The actual running time of the movie is 108 minutes. However, the station has to include 35 × 30-second advertisements and 8 × 15-second station promotions during the movie. The movie cannot run past 10.30 pm as the station then crosses live to an international tennis tournament.

a What length of movie time will need to be cut out to accommodate all the advertising and TV station promotions?

b What percentage of the total movie time is this?

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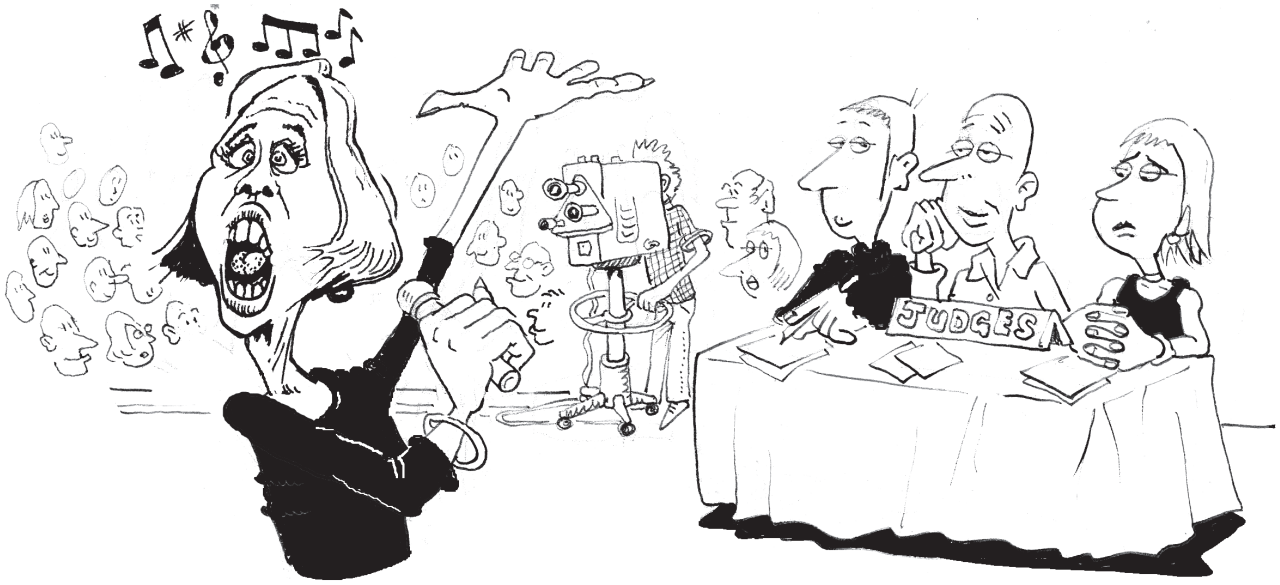
'National Icon'

7

Young talent reaching for the stars, and behind the lights and cameras there's maths including ratio, proportion, transposing equations and applying formulae.

'National Icon' takes young performers and sets them on the road to stardom through a series of performances that are scrutinised carefully by a panel of three judges.



Unless otherwise indicated, express answers correct to 2 decimal places.

1 When auditions were announced for the 'National Icon' competition to be screened on TV, there was enormous interest. Here are the total numbers of initial applicants, broken down by state or territory and gender.

State/Territory	Females	Males
NSW	254	282
Vic.	310	244
Qld	209	187
WA	186	203
SA	177	135
Tas.	64	44
ACT	45	18
NT	28	14

a Calculate the total number of applicants from each state or territory.

b Calculate the total number of applicants nationally.

c Calculate the total number of females nationally.

d Calculate the total number of males nationally.

2 **a** In simplest terms, what was the ratio of females to males nationally?

b In which state or territory was the ratio of females to males greatest?

c In which state or territory was the ratio of females to males lowest?

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'National Icon' (cont'd)

3 An audition list of 200 was finalised from the total number of applicants. What proportion of the total applicants reached the audition list? (Answer to 3 decimal places.)

4 What percentage of the total number of applicants did not reach the audition list? (Answer to 1 decimal place.)

5 What was the probability of any individual applicant making the audition list?

6 The audition list of 200 included 119 females and 81 males. Compare the ratio of females to males on the audition list with the ratio of females to males among the initial applicants. (Use your answer to Question 2a to help you.)

7 Eventually, 12 talented hopefuls were selected as finalists for 'National Icon'. What percentage of the total number of initial applicants does this represent?

8 During the competition, a panel of three people judged the performances and awarded points. This was done by applying two separate algebraic formulae:

- $P = d + p$, where P = performance score
  d = delivery (how well item was sung, scored out of 20)
  p = presentation (of artist during performance, out of 20)

- $S = P \times c$, where S = total score
  P = performance score (out of 40)
  c = complexity (i.e. degree of difficulty of chosen performed piece, ranges 1–3)

a Transpose the first formula to make d the subject ($d =$). Now make p the subject ($p =$).

b Transpose the second formula to make P the subject. Now make c the subject.

9 Using the formulae above, complete the table below and identify which performer won this round of the 'National Icon' competition.

The figures for d and p are averages of the three judges' scores. Answer to 1 decimal place.

Performer	*d*	*p*	*P*	*c*	*S*
Grace	12.5	16.0		1.8	
Felix	14.0		32.5	2.6	
Bryant		14.5	31.0		68.2
Stacey	12.0	16.5		2.3	
Giovanni	16.0		33.5		90.5
Rachel		15.0	29.5	1.7	
Colin	14.5	10.0			63.7
Ebony		11.5	23.0		57.5
Fatima	16.0	13.5		2.4	
Matthew	18.5		37.5	1.9	
Lucien		17.5	30.0		57.0
Holly	11.5	9.5		2.8	

10 Using your results from the table above, rank the contestants in order from highest total score to lowest total score.

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'Out in the Wild'

8

The documentary crews are roaming Africa to find out what some of the biggest, cutest and most striking animals on the planet are up to. They've found some amazing statistics about weights, heights and more.

You can travel to the ends of the Earth … and you don't have to leave home to do it. 'Out in the Wild' brings the world of wildlife to your TV …

1 The 'Out in the Wild' crew visits a region in Tanzania to observe the behaviour of some zebra families. Each zebra family has one male (stallion) and 2–6 females (mares), together with any young foals. There are also bachelor groups of males that do not have families. In the region visited by the crew, the total number of zebras was 216.

a 151 of the zebras lived in families. What was the total number of male 'bachelors'?

b There were 32 stallions in families. How many families were in the group?

c There were a total of 18 foals in the group of zebras. How many adult females were in the group?

d What was the average number of females per family in the group of zebras? (Answer to 1 decimal place.)

2 Further south in Africa, in Botswana, another 'Out in the Wild' crew visits a meerkat colony with 24 meerkats living together. The total length of an adult meerkat is 25–31 cm, and their tail length is typically 19–24 cm. They weigh 620–940 g.

a In this meerkat colony there are 14 males and 10 females. What is the ratio of males to females?

b The heaviest male meerkat in the colony is 830 g and the heaviest female is 770 g. What proportion of the male's weight is the female's weight? (Answer to 2 decimal places.)

c Four of the juvenile meerkats have a combined tail length of 52 cm. What is their average tail length?

d One juvenile has a tail 12.4 cm long. When fully grown, her tail will measure 21.2 cm. What percentage of the fully-grown tail length is her tail now? (Answer to 1 decimal place.)

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3 One of the most popular segments on 'Out in the Wild' was the filming of the birth of a baby giraffe in the wild, six years ago.

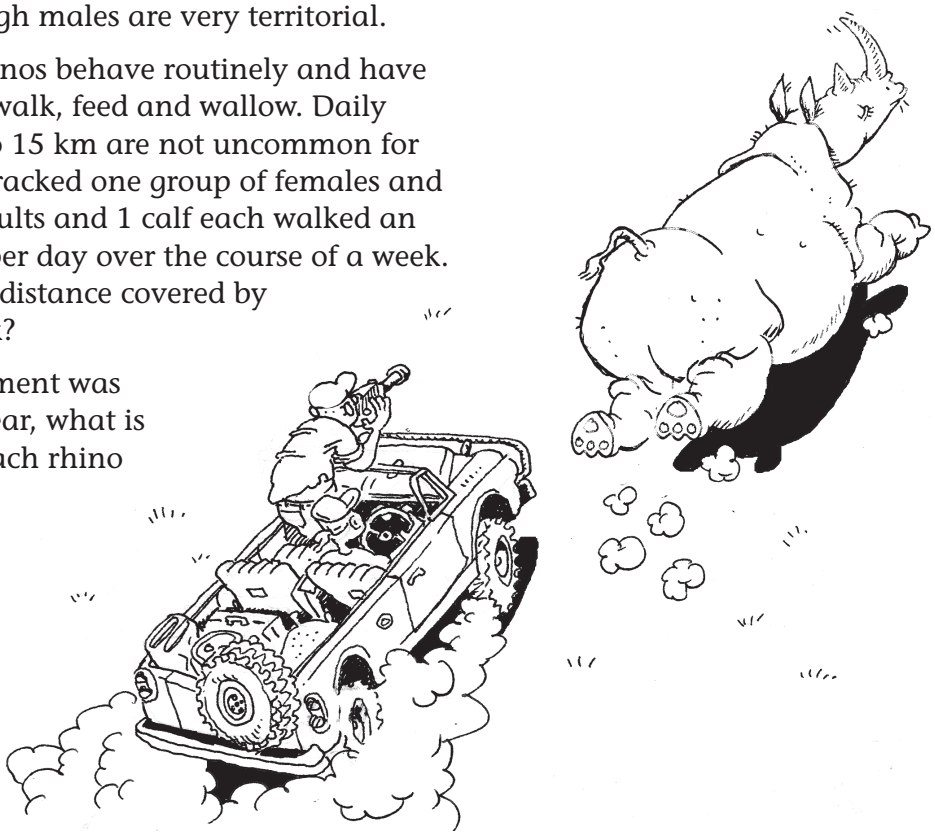
a Female giraffes gestate or carry their young for 457 days. How many months is this? (Answer to the nearest whole month.)

b When it was born, the young male calf weighed 61 kg and stood 2.01 m high. Six years later, the crew revisits the area just south of the Sahara Desert and catches up with the same giraffe. Now he is an adult bull weighing 1345 kg and standing 5.67 m high. When the calf was born, what percentage of its adult weight and height were its birth weight and height? (Answer to 1 decimal place.)

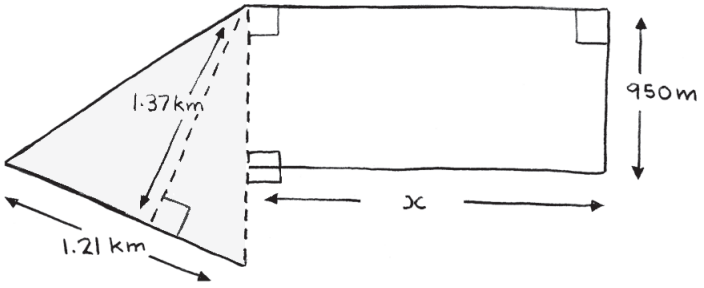
4 A recent 2-hour special of 'Out in the Wild' was filmed at Kruger National Park in South Africa. Part of it covered the habits of a population of Southern White Rhinoceros. These animals generally move in small groups and are quite sociable, although males are very territorial.

a Southern White Rhinos behave routinely and have favourite places to walk, feed and wallow. Daily movements of up to 15 km are not uncommon for females. The crew tracked one group of females and found that the 4 adults and 1 calf each walked an average of 7.8 km per day over the course of a week. What was the total distance covered by one rhino in a week?

b If this rate of movement was maintained for a year, what is the total distance each rhino will have covered?



c One bull rhino spent the entire seven days in his home territory, an area of savannah totalling 2.37 km^2, in the shape shown below. Calculate the value of x, to 2 decimal places).



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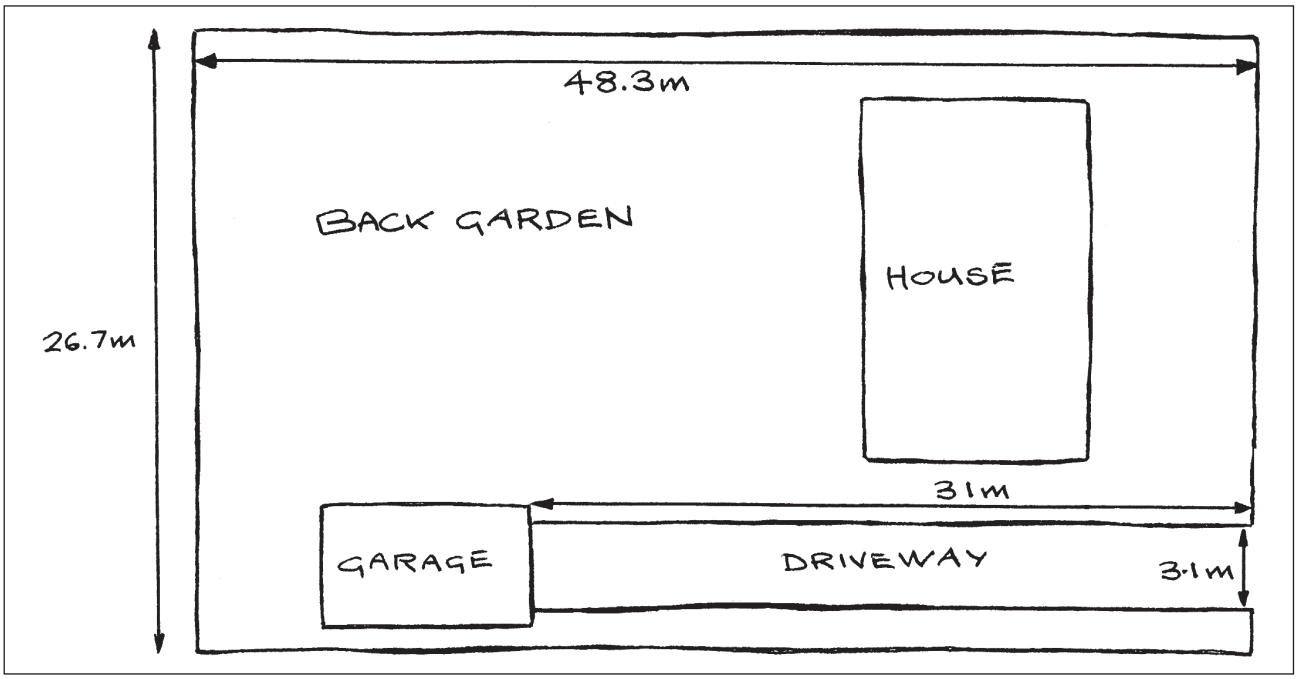
'Backyard Bonanza'

Transforming gardens – the team goes beserk on geometric shapes, costing and timing jobs, and area calculations.

In the hit TV program 'Backyard Bonanza' Jack Dare is the star of the show. He and his team of garden landscapers, carpenters and horticulturalists visit someone's home and transform the garden over the course of a weekend.

This is the 'before' plan that Jack drew up for a property they visited recently.



Not to scale

Unless otherwise indicated, express answers correct to 2 decimal places.

1 What is the total area of this property?

2 The team paved the driveway with pavers. What total area was paved?

3 The top of each paver measured 19 cm × 42 cm. How many were needed to complete the driveway? (Round off to the next whole paver.)

4 Jack can lay 42 pavers in one hour. If his two assistants can keep up with him, how long would it take the three of them to pave the entire driveway? (Express your answer in hours and minutes.)

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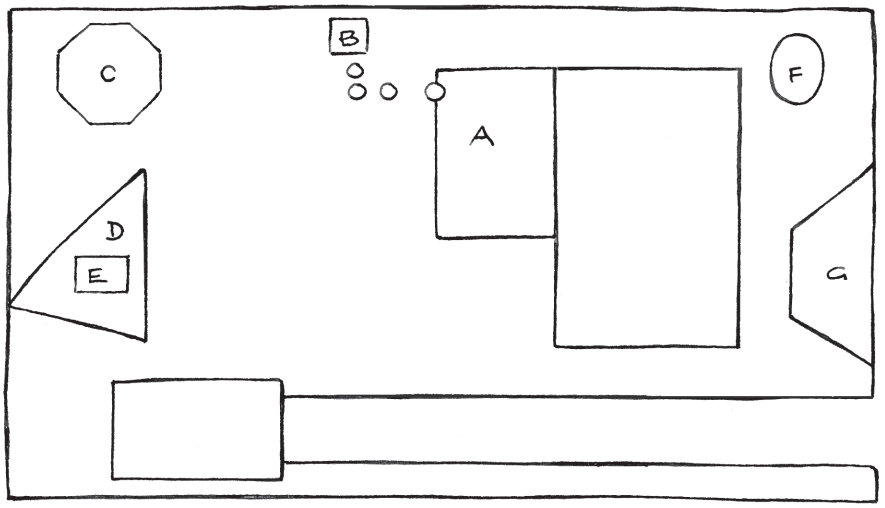
At the end of the weekend, Jack drew this diagram to show how the property looked with its new features:



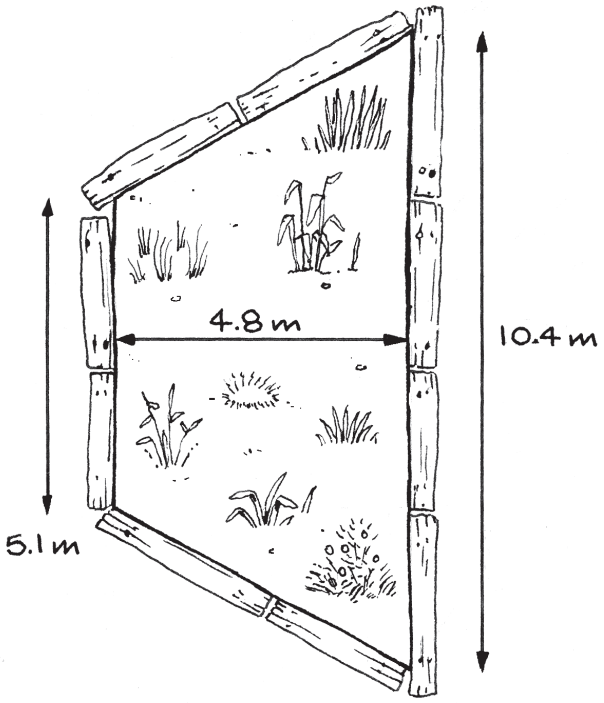
A - pergola
B - permanent BBQ (with paving stones)
C - gazebo
D - shade sail
E - children's sandpit
F - fish pond
G - bordered native grass plantation

Not to scale

5 Identify the shape of each new feature in the garden. (A to G)

6 The pergola has a total of 9 upright support posts, each measuring 2.4 m × 0.1 m × 0.1 m. The cost of timber used for the support posts is $2.35 per metre length. What is the total cost of the support posts?

7 The borders around the native grass plantation are made from sleepers. To prepare the area, the soil was broken up and aerated, and organic matter was added. Eight different native grass species were then planted and the area was mulched to maximise water retention and to control weed growth. The dimensions of the plantation are shown on the diagram.



a Calculate the area of the plantation. (Express your answer correct to 1 decimal place.)

b Mulch was uniformly added to a depth of 5 cm. Calculate the volume of mulch used. (Express your answer in cubic metres to 1 decimal place.)

c The mulch cost $\$38/m^3$ + GST. (GST is an extra 10%.) What was the total cost of the mulch used?

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Here is the News

10

Breaking stories about sport, natural disasters, traffic, crime, the lottery and weather all bring their own set of mathematical challenges. Join us after this short break for a closer look at percentages, proportions, costs, averages and more.

Crime, wars, natural disasters, politics, sport, entertainment and weather – news reporters venture far and wide to make sure you're kept informed about what's happening and who it's happening to.

Unless otherwise indicated, express answers correct to 2 decimal places.

1 The police commissioner announced crime figures for the past year. She reported that the number of home burglaries had decreased by 439 from last year, with the number of reported burglaries this year totaling 4891.

a How many burglaries were reported last year?

b By what percentage had the number of burglaries decreased? Answer correct to 1 decimal place.

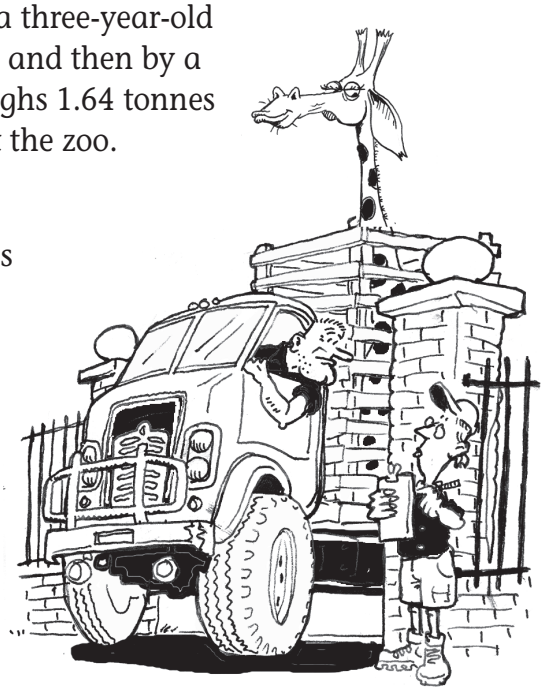
2 Traffic came to a standstill in the city today when a truck carrying a load of bananas overturned. Fortunately no one was injured, but two-fifths of a tonne of bananas were ruined. If bananas retail at $8.99 a kilogram, what is the value of the bananas that were ruined?

3 The Prime Minister said funding would be made available this year to extend Highway 1 by 34 kilometres. The total funding required is $182 million.

a What is the cost per kilometre of new road?

b At this rate, how much would it cost to extend the highway a total of 51 kilometres?

4 The city zoo took delivery of its newest attraction: a three-year-old male giraffe. It was transported by ship to the port and then by a specially prepared truck to the zoo. The giraffe weighs 1.64 tonnes and stands 3.68 metres high. It joins four others at the zoo.



a How many kilograms does the giraffe weigh?

b If the combined weight of the zoo's five giraffes is 5.87 tonnes, what is the average weight of each one in kilograms?

c How many centimetres high is the giraffe?

The giraffe is not the heaviest animal at the zoo. One of the elephants weighs 4010 kilograms.

d What is the elephant's weight in tonnes?

e In tonnes, how much heavier is the elephant than the giraffe?

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5 The Rockets re-signed their star defender today in a deal believed to be worth a total of $1.6 million over five years. The contract is structured so he will be paid 0.12 of his fee in the first year, 0.13 in the second year, and 0.22 in the third and fourth years.

a What proportion of the total fee does he get paid in the fifth year?

b Calculate how much money he gets paid in each of the five years of his contract.

6 Overseas, and an earthquake has struck an isolated part of Turkey. Scientists report it measured 6.0 on the Richter scale. The Richter scale is a logarithmic scale that measures the size (amplitude) of the shockwave resulting from an earthquake. Each whole number increase in the Richter scale means the size of the shockwave increases ten times.

a How much bigger are the shockwaves of this earthquake than one that measures 2.0 on the Richter scale?

For each whole number increase on the Richter scale, approximately 31 times more destructive energy is released by the earthquake.

b How much more energy would be released by this earthquake in Turkey than one measuring 3.0 on the Richter scale?

On average, about 120 earthquakes of the size of the one in Turkey occur around the world each year.

c Complete the sentence: This means that one such earthquake occurs on average every _____ days in a year.

7 In the mid-week lottery there are three divisions. The first division prize of $2.658 million was shared between 13 winners. The second division prize of $65 433 was shared between 43 winners, and the third division prize pool of $11 987 was shared between 104 winners. Answer these questions to the nearest dollar.

a What was the total prize pool?

b How much money did each of the first division winners receive?

c How much money did each of the second division winners receive?

d How much money did each of the third division winners receive?

e The money for the lottery comes from ticket sales. If the government takes 20% of the money collected from ticket sales before any prizes are paid out, what was the total value of ticket sales in the lottery?

f How much money did the government take out?

8 Finally, the weather. Tomorrow's minimum temperature is forecast to be 13° Celsius, and the maximum 24° Celsius.

Convert tomorrow's minimum and maximum temperatures into degrees Fahrenheit. Answer correctly to one decimal place.

To convert degrees Celsius to degrees Fahrenheit, use this formula:

$$TF = (\frac{9}{5} \times TC) + 32$$

TF = temperature in degrees Fahrenheit (°F)

TC = temperature in degrees Celsius (°C)

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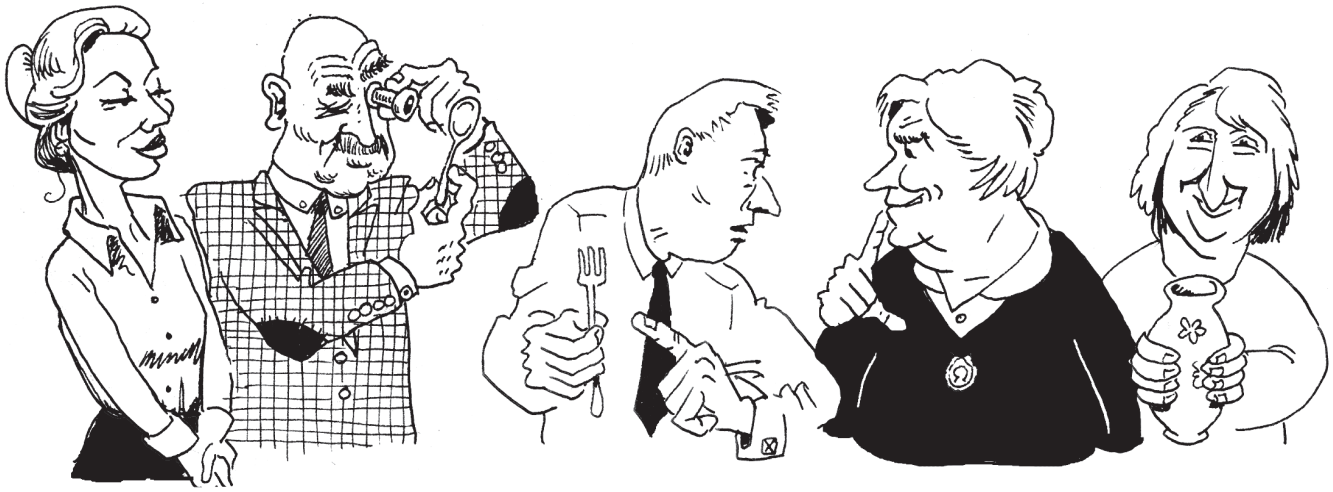
'What's it Worth?'

11

The team casts its expert eye over various valuables, and tells us exactly what they think they're worth. It's all about buying price and selling price, and how they change over the years.

Based on a British show, 'What's it Worth' features a team of antique and collectables experts who visit shows and exhibitions around the country. The experts look at various objects owned by different people and value these objects. Sometimes the owners get a very nice surprise … other times, not so nice!



1 A pair of vases from the 1920s is valued at $950 for the two. The owners had purchased them in England five years ago for £120 each. If the exchange rate is one English pound for $2.40 Australian, how much more is the pair of vases worth now than when purchased? (Answer in $ Australian.)

2 The 'What's it Worth' experts value a 200-year-old Chinese statue at $28 750. It is made of solid gold and weighs 835 g. If gold is worth $790 Australian per ounce, and there are 28 grams to an ounce, calculate the value of the gold in the statue (to the nearest whole dollar).

3 An unusual and rare set of rice bowls featuring handpainted Japanese artwork dating back to 1907 was presented for valuation.

a The original owner of the bowls bought them new, 11 years before they were painted. In what year were the bowls made?

b If they were sold as individual bowls, the experts estimated they would be worth up to $95 each. But sold as a complete set, they are worth up to 55% more than the combined cost of the individual pieces. Up to what price could the current owner expect to receive for them if they chose to sell them as a set of six bowls?

c Suggest why the bowls might be worth more as a set.

4 A fanatical collector of soft drink containers started collecting on his 19th birthday. He has just turned 62.

a Over the first 5 years he was collecting, his collection grew from zero to 2360 cans and bottles. What was the average number of containers he collected per year during this time?

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b Over the next 22 years, this average figure decreased to 227 containers collected every year. Over the 3 years after that, he did not add to his collection. Then he collected avidly for the following 4 years, adding another 1929 containers to the collection.

Since then, he has averaged 97 containers per year. If he has never sold or given away any containers, what is the total number of containers in his collection now?

5 An 80-year-old solid blackwood dresser cabinet is auctioned. Its reserve price is set at $1800.

a What is meant by the reserve price?

b Bidding started slowly but became keenly contested by two bidders. The auctioneer started taking bids at $1200. There were two $100 bids, followed by seven $50 bids. The bidding then increased by $20, and there was a total of six such bids. These were followed by three $10 bids, and finally five $5 bids.

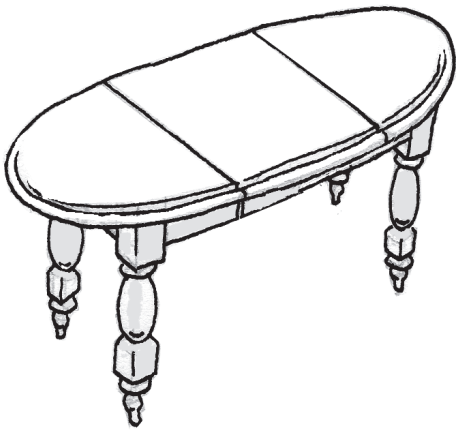
What was the final price bid for the dresser?

c Did the dresser sell at auction? (That is, was it above or below the reserve price?)

d By how much did the dresser exceed or fall short of its reserve price?

e What percentage above or below the reserve price was the final bid?

6 An antique extendable table was valued by the experts at $6500. It was a round table with an insert that slotted in to increase the size of the tabletop. The dimensions of the tabletop and insert are shown.

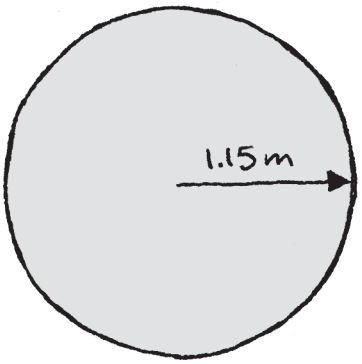


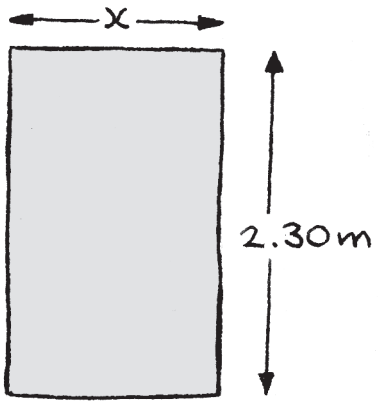
Give answers to 2 decimal places.

a What is the area of the round tabletop?

b The insert increases the area of the tabletop by 1.84 m^2. What is the total area of the extended tabletop with the insert in place?

c Calculate x.





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'Funny Videos Shot at Home'

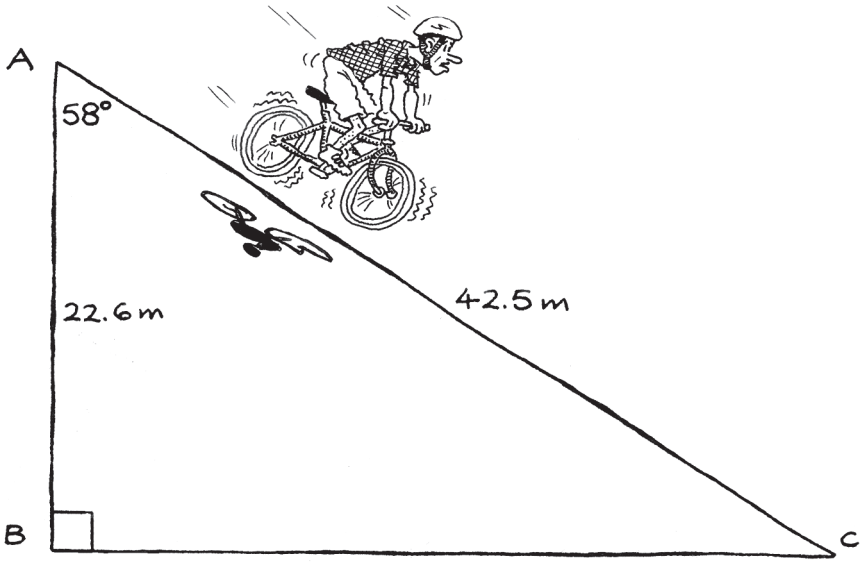
12

The viewers' own blooper reels – join us for a focus on geometry as misfortune is captured by amateur cams.

The thrills, spills and hilarity of those many embarrassing moments that take place every day – just make sure someone's got the video camera handy at the time and the whole country ends up sharing it with you …

Unless otherwise indicated, express answers correct to 2 decimal places.

1 How's this for wheel good fun? When the Jones family went bike riding, Katie got some video of her dad riding down a hill on a bicycle – when the wheel fell off, so did her dad … ouch!



a What is the angle of incline of this hill?

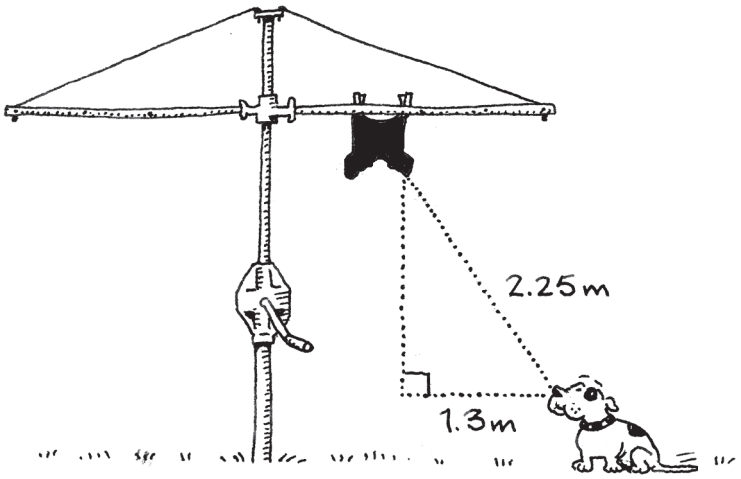
b Katie's dad began from a standing start (0 m/s) at A. He reached C in 6.8 s, when the wheel fell off. What was Dad's average speed in m/s between A and C until he lost the wheel?

c Calculate the horizontal distance (BC) the bike travelled. Answer correctly to 1 decimal place.

2 Uh-oh! Dad's put the washing on the line just when Blackie the dog is thinking it's time for a snack … hmm … Dad's best T-shirt looks good …

a What vertical height did the dog reach when it took the T-shirt from the line?

b If Blackie stands 0.41 m from the ground when she's on all fours, how many times her own height did Blackie jump vertically when she took the T-shirt?



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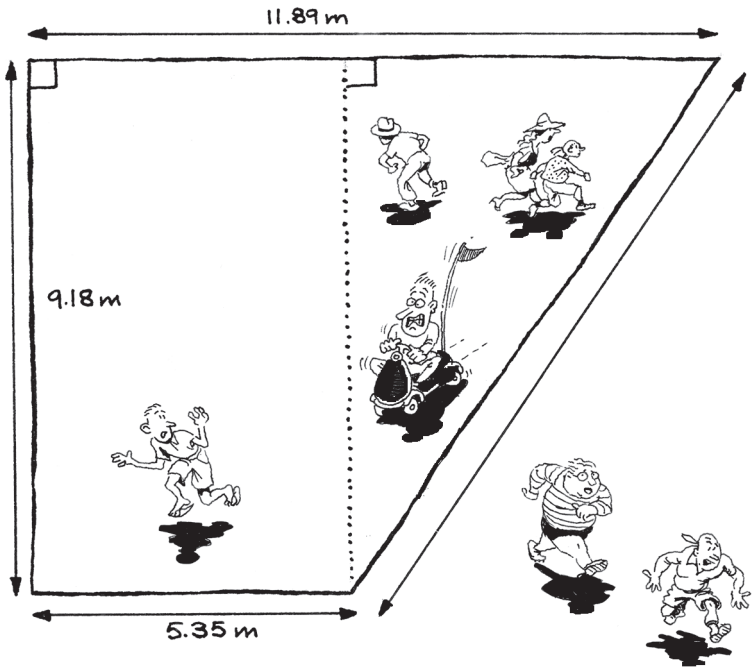
3 Not much happening at this family barbecue in the park until an intrepid teenager decides he'll take Grandma's electric scooter for a spin … when Grandma wasn't on it of course. But how do you stop this thing … oh no!

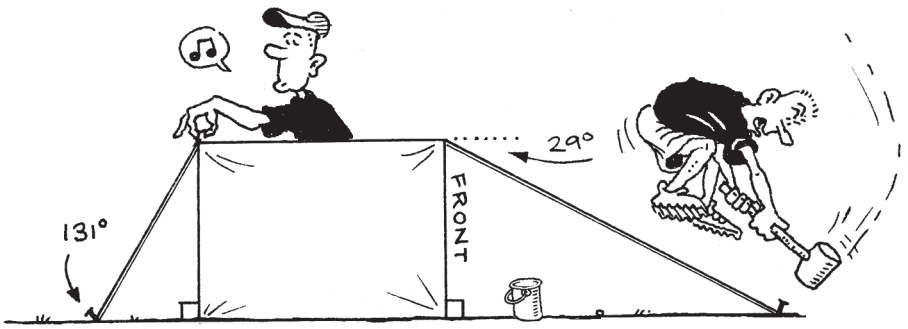


This diagram shows the route taken by the teenager. He sent six people running for their lives and interrupted another family's soccer game before bringing it to a stop. Calculate:

a the area inside the shape of the route taken by the teenager on the electric scooter.

b the total distance travelled once around this route. (Hint – first you'll need to calculate the length of the hypoteneuse of the triangle shown in the diagram).

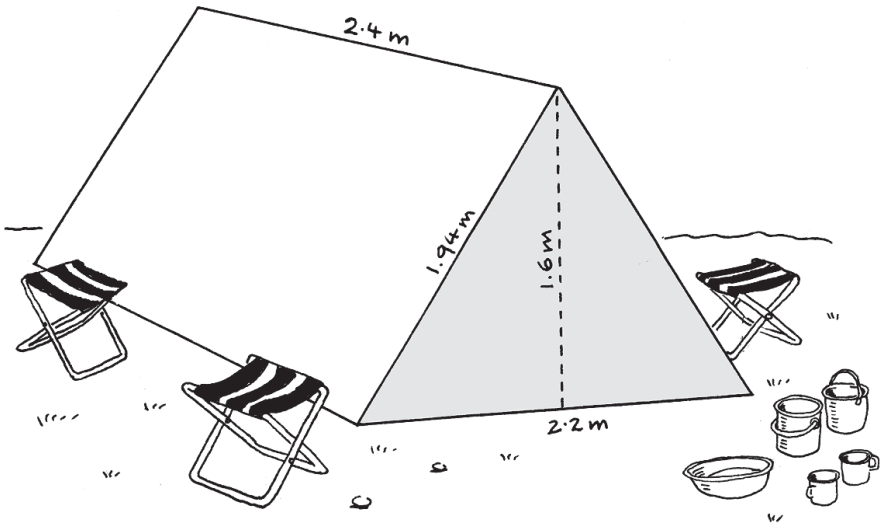
4 Hey, nice job of putting the tent up fellas! Trouble is, it might have been an idea to read the instructions. The front guy rope should make an angle with the ground of 35°–45°. The rear guy rope should make an angle with the ground of 45°–55°. How did our guys go with the guy rope angles? How close were they at each end?



Well, needless to say, the tent soon fell down … our heroes were inside at the time, and they turned into human tent poles.

5 **a** Calculate the total area of high-tech fabric used in making this tent, and remember it has a floor!

b If this fabric costs $5.15 per square metre to purchase, what is the value of the fabric in this tent?



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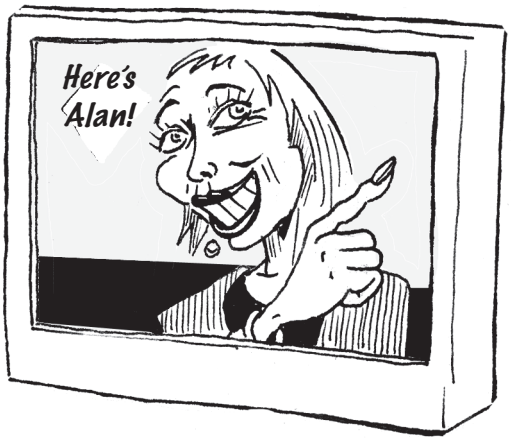
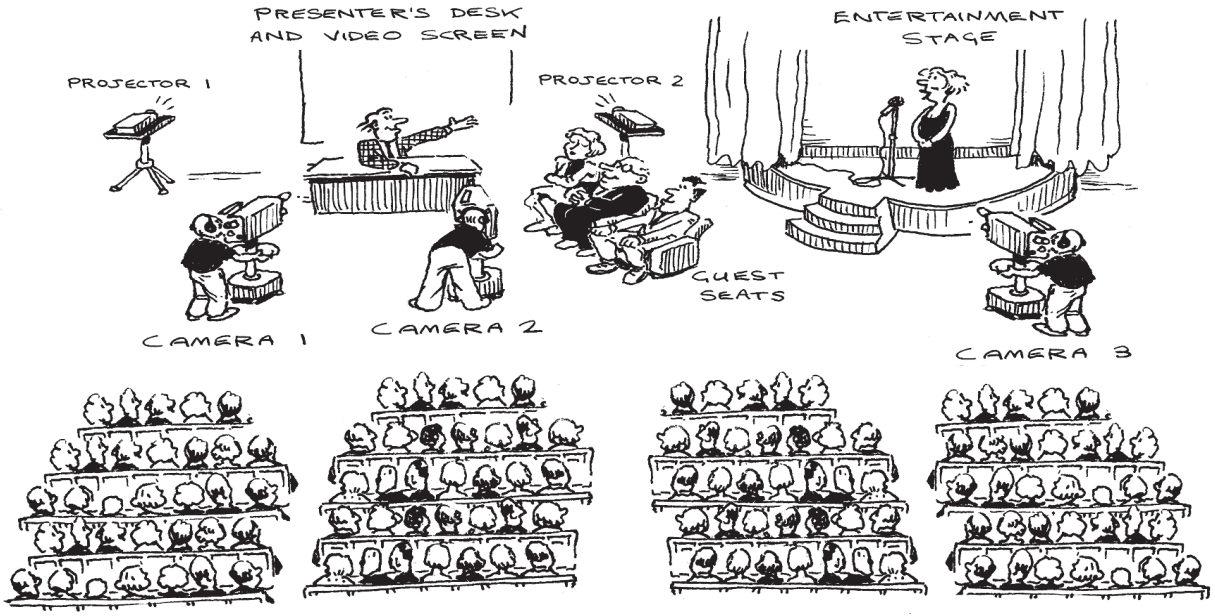
The tonight show host you have when you don't have a tonight show host – a CGI with his very own variety program presents problems about studio angles and show segment timing.

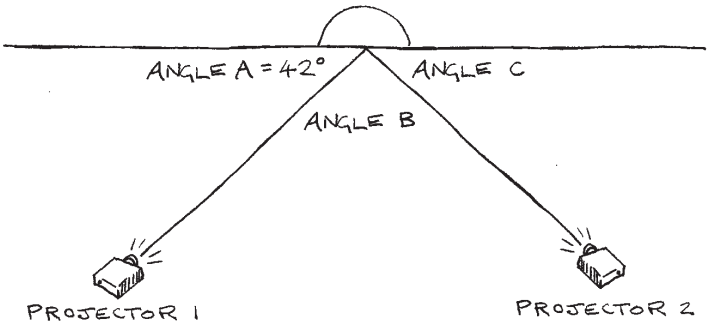


Alan Forsyth is not a real person. He is a CGI (computer-generated image) that plays the role of a tonight show host. Real people feature on his show, which includes interviews, entertainment and competitions that take place in a specially constructed Channel 5 studio, shown below.

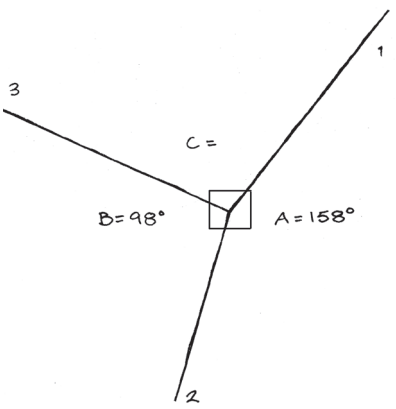
Unless otherwise indicated, express answers correct to 2 decimal places.



1 If the projected image of Alan Forsyth is made by the two projectors shown, and angles A and C are equal, calculate the angle formed between the projector beams as they hit the presenter's desk (i.e. angle B).



2 Camera 2 is filming a comedian on the entertainment stage (1), moves clockwise to pick up a shot of someone in the studio audience (2), then moves clockwise again to film Alan Forsyth's desk (3). Calculate Angle C.



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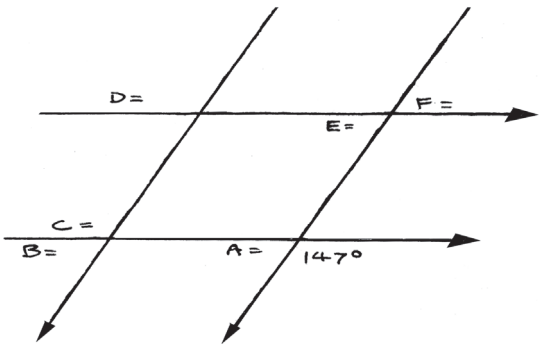
3 The diagram below is a section of the lighting grid placed just below the ceiling of the studio.



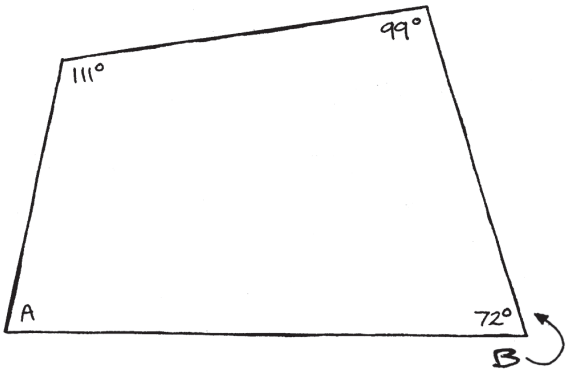
a Fill in the measurements of all the indicated angles, A to F.

b From the diagram, which of the two marked angles are vertically opposite?

c From the diagram, which of the two marked angles are supplementary?

4 **a** This is the entertainment stage. What shape is it?

b State the value of the angles A and B.



5 The maximum size of the studio audience is 132. If all seats are filled and there is a ratio of 6 : 5, females : males. How many of each gender are present in the audience?

6 During one show, each studio guest appears for 6 minutes. The first studio guest is asked 11 questions, the second studio guest is asked 14 questions and the third studio guest 8 questions.

a What is the total number of questions asked?

b What is the average number of questions asked each guest?

c For each guest, calculate the average number of questions asked per minute, correct to 2 decimal places.

7 During the one-hour show, there are the studio guests (each appearing for 6 min), three entertainment items (a band that plays one song lasting 3 min 48 s and another 3 min 17 s, and a comedian whose routine lasts 4 min 11 s); a competition that runs for 5 min 16 s; and 8 min 51 s spread throughout the program of 'Alan Forsyth Live'. The opening credits last for 37 s and the closing credits for 60 s. The remainder of the time is taken up by advertisements.

a What is the total time in minutes taken up by advertisements during the hour?

b What percentage of the one-hour time slot is this?

c There are five advertisement breaks through the show, all of the same length. How long is each ad break?

d If every advertisement shown lasts 30 s, how many advertisements are in each break?

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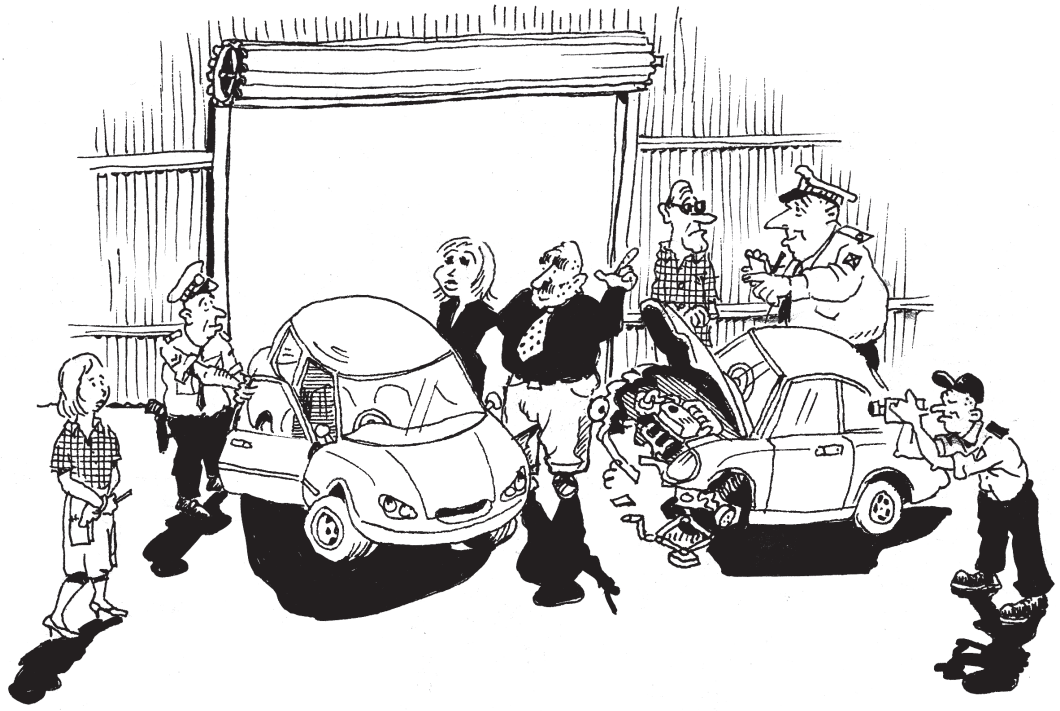
'Cops on the Beat'

14

Forget watching people acting as cops – this is the real stuff! TV crews are dispatched with teams of police as they crunch the numbers on crime with percentages, money, length and graphing.

This action is definitely not acted. 'Cops on the Beat' takes you out on the streets, into homes, shops, offices – even down to the waterfront – where every day and night our police are confronted with a side of life that most of us only ever imagine.



1 'Cops on the Beat' went with a police team to investigate a suspected stolen car operation in the inner city. On arriving at a warehouse, they arrested four men and two women and impounded 14 vehicles that had been reported stolen.

a The police said the vehicles had an estimated total street value of $290 000. Included in the haul was a Porsche worth $105 000. What was the average value of each of the other vehicles found in the warehouse (to the nearest whole dollar)?

b One of the vehicles had some damage to its right-hand side, which had been caused during the theft. The vehicle had an estimated street value of between $16 500 and $17 000 but the police said that if it had been sold illegally with the damage, the asking price would drop by 30–40%. Using these estimations given by the police, calculate the minimum and maximum value of the vehicle had the thieves been able to sell it in its damaged state.

c This operation was the third of its type that this police investigation unit had uncovered in 12 months. The combined total value of all stolen vehicles recovered in the three investigations was $1 389 543, but the total value of all vehicles stolen during those 12 months was estimated at $12 500 000. What percentage of the estimated stolen vehicle market has this investigation unit been able to recover over the past year (to 1 decimal place)?

2 Another investigation followed by 'Cops on the Beat' was into a fraud carried out by the manager of a company. Over the course of two years, she had siphoned off a total of $567 000 from customers' accounts into her own bank accounts for her personal use.

a In one account she had operated for nine months, she was collecting 7.2% interest per annum. She opened this account with $245 000 and did not add to it. How much was in this account when she was caught?

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b From one customer, she had regularly taken an amount of $17.23 every fortnight. She had done this between 1 March and 5 December in the same year. How much had she stolen from this customer in total?

c Of the total $567 000 stolen, she claimed she had lost $98 000 gambling. What percentage of the total stolen funds is this (to 1 decimal place)?

d The police, however, stated that in all 38% of the total amount stolen was unaccounted for. How much money is this?

e Given the woman's confession about the gambling money, how much more money over and above this figure could not be traced?

3 Down to the waterfront – and 'Cops on the Beat' are stopping recreational fishing boats to check the size of the catch. Large quantities of undersized fish are finding their way onto the black market – in particular, King George whiting. Where the police are swooping, the legal maximum limit of King George whiting for recreational fishers is 75, and each fish must be 30 cm or more in length. During the operation police inspected 68 fishing boats. These boats contained a total of 2940 fish of which 1371 were King George whiting. In all, 160 whiting were found to be undersized. Seven of the boats had more than 75 whiting on board.

a What was the minimum number of whiting in total that these seven boats could have contained?

b The seven boats actually contained 1098 whiting between them. What was the average number of whiting on board each of those seven boats (to the nearest whole fish)?

c The seven boats also accounted for 85% of all the undersized whiting seized. How many fish is this?

d Do you think there is evidence here that the owners of these seven boats could be involved in the illegal sale of whiting? Explain.

4 One man was caught with 16 King George whiting in all, including two that were 29.7 cm and 29.85 cm respectively. The man showed police an old battered steel ruler and claimed he used it to measure every fish he caught. The police let him off with a caution.

a By what percentage was each of the undersized fish below the legal limit?

b Do you think the police treated the man fairly? Why or why not?

c What advice might the police have given the man when they cautioned him?

5 The break down of fish inspected during the operation is below. Construct either a column graph or a pie chart to provide a graphic representation of the break down of this data.

King George whiting.....1371
Snapper.....677
Flathead.....319
Barracouta.....171
Morwong.....118
Mullet.....107
Snook.....94
Flounder.....68
Mako shark.....9
Southern bluefin tuna.....6

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'Time Off!'

15

Time for a holiday away – presenters in three beautiful locations pose problems about travel times, costs of holidays, distances, fuel consumption and world time zones.

'Time Off!' takes viewers to a host of different holiday and weekend break destinations in Australia and overseas.

Unless otherwise indicated, express answers correct to 2 decimal places.

1 Presenter Carrie took off to check out a new resort in Port Douglas in Queensland. Port Douglas is 62 km north of Cairns, which is 2500 km from Sydney. Carrie flew from Sydney to Cairns in 2 h 40 min, then travelled by bus from Cairns to Port Douglas, which took 45 min.

a What was her total travelling time?

b What was the total distance she travelled?

c What was the average speed of the aeroplane in km/h?

d What was the average speed of the bus in km/h? Answer to 1 decimal place.

e What was her overall average speed on the entire journey? Answer to 1 decimal place.

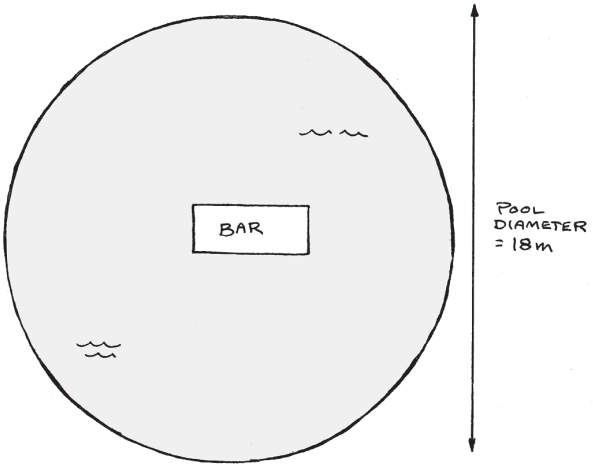
2 At the resort is a large circular swimming pool with a floating bar in the centre, as shown.

a What is the area of the bottom of the pool?

b If the bar is 5 m long × 1.5 m wide, what is the area of the visible surface of the pool?

c What is the circumference of the pool?

d The pool has a uniform depth of 1.5 m. What volume of water does it hold? Answer in m^3 and convert to litres.



3 The resort charges $195 for a double room (including breakfast) per night. For every five nights booked, guests are offered one night free. How much would a couple pay for 15 nights at the resort?

4 The first 50 callers to 'Time Off!' were offered the opportunity to book a double room for $80 per night. What percentage of the usual price is the special offer?

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5 Presenter Wayne went on a guided outback camping trail in a four-wheel drive.

a The four-wheel drive had a fuel tank that held 120 litres, plus a reserve tank of 40 litres. On the week-long trail, they used 2.5 regular tanks of fuel and half the reserve tank. How much fuel did they use altogether?

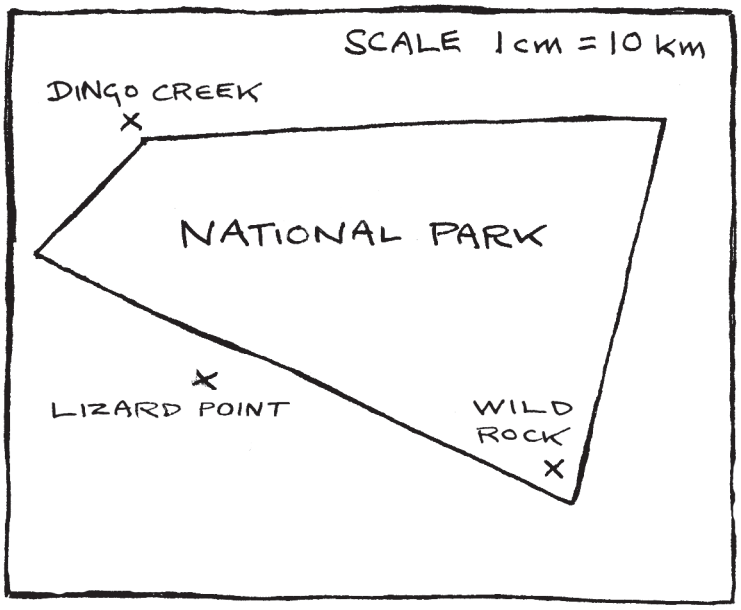
b If fuel was an average of $1.75 a litre, how much did fuel cost over the whole trip?

c The average consumption by the vehicle was 11.5 litres per 100 kilometres over the whole trip. How far did they travel?

6 **a** One night they camped in a national park. The camping site was 45 km from Dingo Creek and 25 km from Lizard Point. Using the map scale and a compass, locate the camping spot.

b Use a protractor to work out the bearing of the camping site from Wild Rock.

c Write the scale of the map as a ratio.



7 Presenter Anna travelled to Malta, a small island in the Mediterranean Sea. Use an atlas to find the latitude and longitude of Malta's capital city Valetta.

8 Malta's time zone is GMT (Greenwich Mean Time) + 1 hour. The east coast of Australia is GMT + 10 hours. When Anna landed in Malta, it was 7.30 pm on a Wednesday. What time and day was it in Sydney?

9 The normal airfare to Malta at the time Anna travelled was $2430 return from Sydney. However, during peak time, July–August in the Northern Hemisphere, there is a 30% loading on the airfare. How much would it have cost to fly return to Malta had she gone in peak time?

10 Tourism has become a significant industry in Malta over the past four or five decades. Recent figures show there are close to 200 different accommodation venues providing a total of 39 105 beds, 48% of which are in four-star and five-star hotels. How many beds are there in accommodation venues that have a rating of less than four stars?

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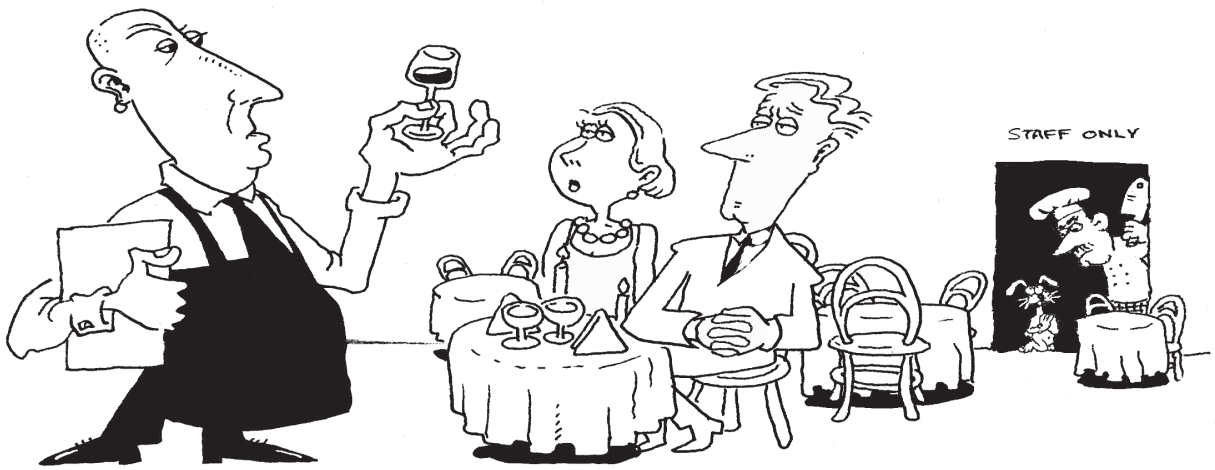
'My Reality Restaurant' 16

The menu's just the start of it – this is the serious end of the business where it's time to work out how the income and costs balance out, with a bit of graphing thrown in for dessert, in the race to get the nod from the judges.



In 'My Reality Restaurant', four couples set up, open and operate a restaurant. On television we follow their progress until, ultimately, one couple's restaurant is voted the best by a panel of judges and the viewing audience – but each couple's ability to run their restaurant as a business is also carefully assessed.

The couples are:

• Alex and Hayley • Rhonda and Issa • Carlos and Mae • Victor and Inez

Unless otherwise indicated, express answers correct to 2 decimal places.

1 Each of the couples projected how many meals they could serve in their first week, and planned their menus and work rosters accordingly.

Couple	Cost of food	Cost of wages (not incl. owners)	Av. cost of meal	Proj. no. of meals	Total income from meals	Gross profit or loss for week	Other running costs	Net profit or loss for week
Alex & Hayley	$1430	$2200	$22.50	290			$960	
Rhonda & Issa	$1890	$1950	$18.50	345			$580	
Carlos & Mae	$2450	$2975	$24.70	400			$1125	
Inez & Victor	$1680	$2100	$17.95	325			$620	

a Complete the **Total income from meals** column:
Income = projected no. × av. cost

b Complete the **Gross profit or loss for the week** column:
Gross profit or loss = Income – cost of (food + wages)

c Complete the **Net profit or loss for the week** column:
Net profit or loss = Gross profit or loss – other running costs

2 **a** Rank the couples in order from highest net profit to lowest net profit.

b Was the order the same for gross profit figures? Explain.

c In the table on the next page, state whether changing each cost factor individually as described would mean an increase or decrease in the net profit.

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Cost factor	Effect on net profit (increase or decrease)
Increase cost of food	
Decrease cost of food	
Increase cost of wages	
Decrease cost of wages	
Increase av. cost of meal	
Decrease av. cost of meal	
Increase projected no. of meals	
Decrease projected no. of meals	
Increase other running costs	
Decrease other running costs	

Alex & Hayley	265
Rhonda & Issa	393
Carlos & Mae	368
Victor & Inez	377

3 Here is the actual number of meals sold in their first week. State by how many meals each couple fell short of, or exceeded, their projected figure.

4 State the percentage error in each couple's projections. Which couple was closest in their projection? Use this equation:

$$\frac{\text{difference between actual meals and projected meals}}{\text{projected meals}} \times 100$$

5 If all other costs remained the same as in the table in Question 1, complete this table by calculating:

a the actual total income for meals

b the actual gross business profit for each couple in that week

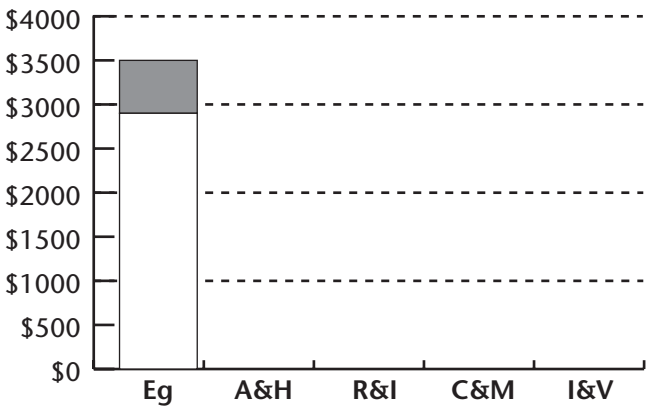
c the actual net business profit for each couple in that week.

Couple	Cost of food	Cost of wages (not incl. owners)	Av. cost of meal	Meals sold	Total income from meals	Gross profit or loss for week	Other running costs	Net profit or loss for week
Alex & Hayley	$1430	$2200	$22.50	265			$960	
Rhonda & Issa	$1890	$1950	$18.50	393			$580	
Carlos & Mae	$2450	$2975	$24.70	368			$1125	
Inez & Victor	$1680	$2100	$17.95	377			$620	

6 Complete a column graph on the axes at right, showing the gross profit and net profit for each couple. Use the information from Question 2.

Use different colours or shading to show the two profit figures. A hypothetical example is given.

Example couple (Eg) have a gross profit of $3500 and a net profit of $2900.



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'Wonderful World of Sports'

17

This is where maths can make or break careers! Figure out the speeds, the scores and the distances on golf courses, Aussie Rules football fields and swimming pools ... not to mention an encounter with Pythagoras on the rugby pitch.

There's entire channels devoted to it – we love our sport, and TV is where it's happening 24/7 … the winners, losers, highs and lows – our own local heroes, and the international stars …

Unless otherwise indicated, express answers correct to 2 decimal places.

1 It's the last round of the Pro-Oz Golf Classic. The leaders' board looks like this after 54 holes (3 × 18 holes).

a Par for the 18-hole course is 73. How many strokes has it taken each of the golfers to complete 54 holes so far?

b For the first 18 holes Jorgensen scored 4 under par (69), then in the second 18 holes she scored 1 over par (74). What was her score on the third 18 holes?

c On the final 18 holes the 10 golfers scored the following: Johnson 76; Gullan 73; Robinson 74; Villani 71; Jorgensen 75; Katazumi 68; Anderson 71; Li 77; Broderick 69; Hansen 73.

How did the leaders' board look at the end of the tournament after 72 holes in total? Fill in the table at the right. Was a play-off required to decide the winner?

Player & nationality	Strokes above/ below par
1 Johnson (Aus)	–13
2 Gullan (India)	–12
3 Robinson (USA)	–12
4 Villani (Italy)	–10
5 Jorgensen (Sweden)	–9
6 Katazumi (Japan)	–9
7 Anderson (Aus)	–8
8 Li (China)	–8
9 Broderick (S. Africa)	–8
10 Hansen (USA)	–7

Player & nationality	Strokes above/ below par
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	

2 In AFL football, the number of possessions a player has is the sum of the number of kicks and handballs he delivers.

The top 5 possession-getters in the match between the South Coast Falcons and the Brisbane Cheetahs, in order from the highest possession-getter down, are listed in the table below right.

Complete the table using the following information:

- Samuels had an identical number of kicks and handballs.
- D'Amor had the same number of handballs as his teammate but 40% fewer kicks.

Player	Kicks	Handballs	Possessions
Stewart (Cheetahs)		12	32
Henderson (Falcons)			
Iskra (Falcons)			28
Samuels (Falcons)			
D'Amor (Cheetahs)			

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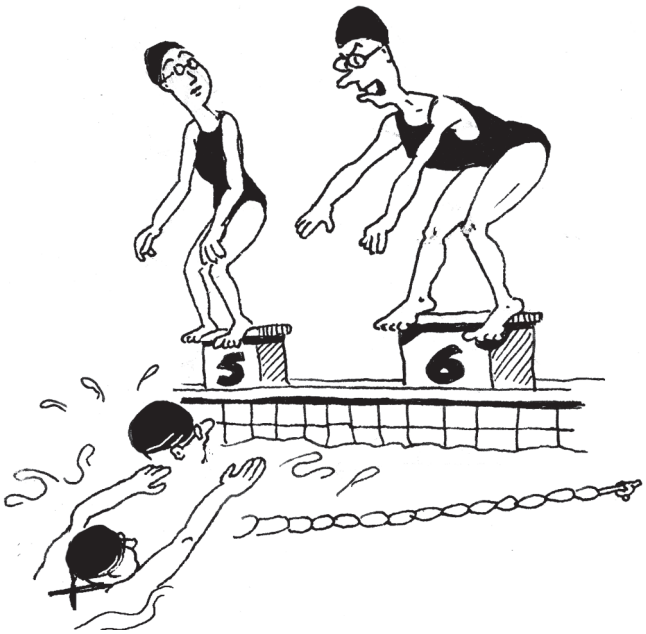
- Henderson had three-fifths of the number of kicks that Stewart had, but 50% more handballs.
- Iskra had three times as many kicks as he did handballs.
- The three top Falcons's players averaged 28 possessions each.

3 The women's medley relay team of Jones, Donekos, O'Donnell and Chang won their 4 × 100-metre event at the world swimming championships in a time of 4.00.27.

a How many laps of the 50-metre pool did the women collectively swim?

b If the world record at the time was 3.57.32, how far outside the record were they?

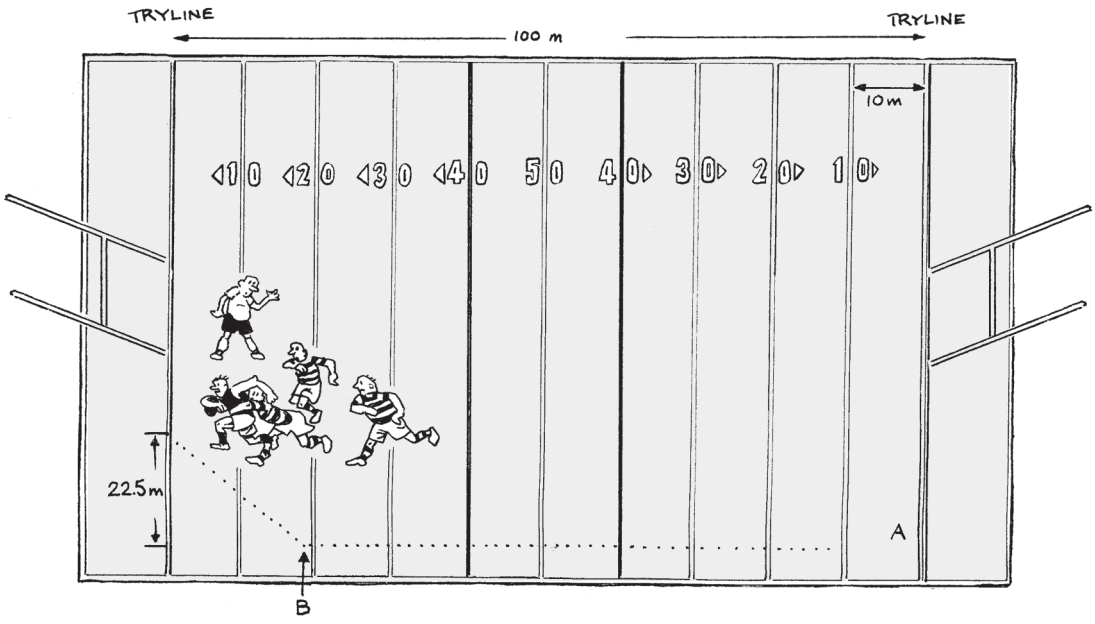
c Chang swam the freestyle in 54.09, O'Donnell the backstroke in 1.01.05, and Donekos the breastroke in 1.07.44. What was Jones's time for the butterfly?



4 In the diagram below, the dotted line represents a run made by the Leopards's winger Steve Turner to score a memorable try in the Rugby League Grand Final. The total distance between the trylines is 100 m, and Turner started his run 11 m in from the tryline the Leopards were defending at Point A.

If Point B is 19.6 m from the tryline the Leopards were attacking, what was the total length of Steve Turner's run?

Hint: think Pythagoras!



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Logging and editing

18

You shoot two hours of tape and end up using 25 seconds of it. Welcome to a behind-the-scenes look at how all that stuff ends up on the cutting room floor, from the hours and minutes down to the hundredths of seconds.

Behind the scenes in any commercial or public television studio are many people who work on editing and preparing finished programs for broadcast. They use a range of technology to process footage shot on film by camera operators. Timing has to be highly accurate, so when camera operators shoot film, in or out of the studio, a time code appears on the tape accurate to hundredths of a second. Editors can then take the material shot by the camera operators and select the bits they want to go into a finished program with a high degree of precision.

To read a time code

Time codes are shown as numbers with colons or points separating them into pairs that show the hours : minutes up to 60 : whole seconds up to 60 : tenths and hundredths of seconds. Here are some examples:

00 : 00 : 13 : 65 = 13.65 seconds

00 : 07 : 22 : 91 = 7 minutes, 22.91 seconds

01 : 34 : 56 : 79 = 1 hour, 34 minutes, 56.79 seconds

1 What do these time codes read?

a 02 : 06 : 17 : 08

b 00 : 28 : 00 : 00

c 00 : 00 : 00 : 04

d 17 : 19 : 45 : 12

2 An editor is working on compiling footage shot in the field on a story for the nightly news bulletin. She uses a grab from an interview that runs between the time codes 00 : 09 : 14 : 00 and 00 : 10 : 02 : 44. Exactly how long did the interview grab run in the news report?

3 An editing team is working on assembling highlights from the previous day's football games to show on a Sunday sports program. Two games were played.

In the table on the next page is the record of tape logging kept by the editing team for the highlights package. It shows the start and finish point from the time code readings on the camera tapes relating to each highlight used for the package.

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Game	Tape no.	Start	Finish	Total (s)
Game 1	1	00 : 04 : 12 : 50	00 : 04 : 19 : 40	
		00 : 11 : 45 : 09	00 : 12 : 18 : 75	
		00 : 33 : 57 : 00	00 : 34 : 18 : 19	
		01 : 09 : 55 : 66	01 : 10 : 09 : 81	
			TOTAL HIGHLIGHTS GAME 1 =	
Game 2	1	00 : 23 : 36 : 47	00 : 23 : 52 : 06	
	2	00 : 07 : 42 : 45	00 : 08 : 00 : 02	
		00 : 48 : 55 : 98	00 : 49 : 11 : 31	
		01 : 17 : 58 : 14	01 : 18 : 01 : 49	
			TOTAL HIGHLIGHTS GAME 2 =	

a Complete all the figures in the Total column. If relevant, when adding up the total highlights from each game, express your answer in minutes and seconds.

b Which game had more footage in the highlights package?

c How much more footage came from this game?

4 The tapes used by a camera crew for shooting a drama series last for exactly 57 minutes of filming time each.

a For one half-hour episode of the drama series, which is actually 22 minutes of program time, the editors were given a total of 3 hours and 54 minutes of footage to cut down. How many tapes had been used to shoot the entire footage (to the nearest whole tape)?

b What percentage of the footage shot by the camera operators actually made it to the finished program that went to air (to 1 decimal place)?

5 An editor needs to insert exactly 19 seconds of footage at the end of a comedy program. He has footage of a short comedy sketch that will fit well, but the time code on the tape mysteriously disappears from it when it reads 00 : 48 : 21 : 78. If the time code reading at the start of the section of tape being used is 00 : 48 : 09 : 11, exactly how much more tape must the editor run in to the program after the time code disappears?

6 A documentary film crew recorded a number of street interviews. Unfortunately, on returning to the studio it was discovered that part of the 120-minute tape was damaged. These sections of tape were unable to be used:

- 00 : 17 : 18 : 95 to 00 : 21 : 43 : 00
- 00 : 29 : 57 : 04 to 00 : 33 : 12 : 38
- 00 : 49 : 38 : 81 to 00 : 55 : 31 : 10
- 01 : 11 : 20 : 42 to 01 : 19 : 44 : 07

a How much of the footage is now unusable?

b The best material was gathered in the first 25 minutes of filming. What percentage of that material is now unable to be used (to 1 decimal place)?

c There was a total of 1 hour, 39 minutes and 17.89 seconds of footage shot by this crew. How much of the total footage is now usable?

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TV ratings 1

19

With all the channels scrambling for market share, keeping on top of who is watching what is big business. The figures are all there – population, sample size and percentages – so you work out what they mean.

Ratings are critical to the operations of commercial television stations. By having independent assessments made of the number of people who watch different programs on different channels at different times each day, commercial TV stations, particularly free-to-air channels, can attract advertisers. In effect, the television channels 'sell their audience' to advertisers.

Ratings data isn't just about how many people watch a program.

It's about a much bigger demographic picture that enables advertisers to choose the best times and locations to run their advertisements. This bigger demographic picture includes the age range of these viewers, whether they are metropolitan or rural residents, their occupations and general income level.

Ratings agencies recruit households to be part of their data collection. Once a household is recruited and some details about its occupants recorded, the monitoring of all TV sets in the household is done electronically and the data processed by a ratings agency.

The following activities are based on a simple model of collecting ratings data. Unless otherwise indicated, express answers correct to 2 decimal places.

1 In statistics, the following terms are frequently used when conducting surveys. Briefly define each one.

- **a** population
- **b** sample
- **c** sample size

2 A ratings agency carries out an analysis of TV viewing separately for Australia's six state capital cities, as well as combining the data from the states to provide a national analysis. It also carries out analysis of TV viewing in rural and remote areas, although the data from the capital cities is the most valuable for TV stations and most advertisers.

- **a** Suggest some reasons why the viewing habits in different capital cities are analysed separately, as well as an analysis being made for national viewing habits.
- **b** Why is the statement made that the data from the capital cities is most valuable for the TV stations and most advertisers?

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3 The TV ratings agency, Oozwatchinwot, carries out a ratings survey in one of Australia's major state capitals. The population of the city is 3.5 million. Survey results indicate that 0.02% of the population never watches any television at all.

Oozwatchinwot has recruited 1500 households from which it collects data. The total number of TV viewers in the 1500 households is 5000 people.

a Describe the population for this survey.

b What is the sample?

c What is the sample size?

d Is the sample size large enough to give a realistic idea of who watches what?

e How many people in this city never watch TV at all?

f Is this figure significant enough to be factored into the survey results?

4 Here are some results from 24 hours during a weekend ratings survey. They relate to the 5000 viewers in the recruited households. Research has shown that at different times of the day, different numbers of viewers are watching, and this is indicated by the percentages underneath the time slots. So between 6 am and 9 am, a maximum of 19% of the viewing audience is likely to be watching TV. Therefore, this means that of the 5000 people in the surveyed households, 950 people would be watching.

Channel	6 am – 9 am 19% viewing (950)	9 am – noon 35% viewing	Noon – 6 pm 52.5% viewing	6 pm – 9 pm 78% viewing	9 pm – midnight 38% viewing	Midnight – 6 am 8% viewing
3	15.2%	12.6%	42.6%	18.8%	21.6%	33.6%
4	14.0%	16.7%	9.1%	21.1%	15.8%	21.1%
5	30.7%	26.4%	24.0%	22.0%	28.7%	22.3%
6	31.8%	22.0%	11.9%	19.3%	21.5%	19.5%
ABC	6.0%	13.3%	7.2%	12.5%	8.8%	3.5%
SBS	2.3%	9.0%	5.2%	6.3%	3.6%	–

a Of the timeslots, which would be considered the prime viewing time? Explain your answer.

b Suggest some reasons why this timeslot would be the prime viewing timeslot.

c How do you think the rates charged to advertisers might vary during the 24-hour period?

d The 24 hours referred to above run from 6 am Saturday to 6 am Sunday. On the Saturday afternoon a highly popular major annual sporting event took place, televised by one channel. This event draws one of the biggest TV audiences for the whole year. Which channel televised it?

e Of the 5000 people who live in the surveyed households, how many were watching this event?

f What do you think might happen with the rates charged to advertisers during this major event?

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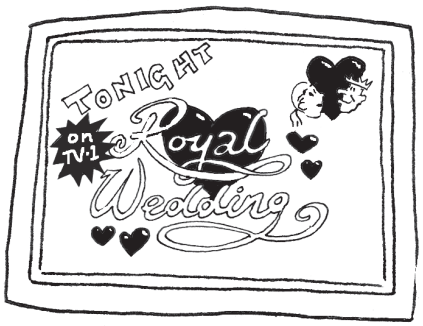
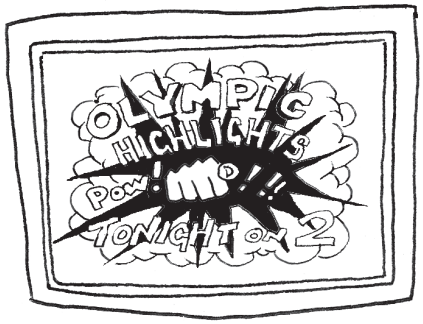
TV ratings 2

20

More about how they watch what we watch – comparing viewers across channels and time slots using more sample sizes and percentages, with some graphing highlights.





As shown in TV ratings 1, these are the results from a 24-hour period during a weekend ratings survey. They relate to 5000 viewers in the recruited households.

Channel	6 am – 9 am 19% viewing (950)	9 am – noon 35% viewing	Noon – 6 pm 52.5% viewing	6 pm – 9 pm 78% viewing	9 pm – midnight 38% viewing	Midnight – 6 am 8% viewing
3	15.2%	12.6%	42.6%	18.8%	21.6%	33.6%
4	14.0%	16.7%	9.1%	21.1%	15.8%	21.1%
5	30.7%	26.4%	24.0%	22.0%	28.7%	22.3%
6	31.8%	22.0%	11.9%	19.3%	21.5%	19.5%
ABC	6.0%	13.3%	7.2%	12.5%	8.8%	3.5%
SBS	2.3%	9.0%	5.2%	6.3%	3.6%	–

1 Calculate the total number of viewers for each timeslot. The 6 am to 9 am timeslot has been calculated for you.

a 6 am–9 am = 19% of the 5000 viewers in the sample = 950 viewers

b 9 am–noon =

c Noon–6 pm =

d 6 pm–9 pm =

e 9 pm–midnight =

f Midnight–6 am =

2 Of the 5000 people who live in the surveyed households, how many were watching (round answers up to nearest whole number):

a SBS between 9 am and noon

b Channel 6 between midnight and 6 am

c Channel 4 between noon and 6 pm

d Channel 5 between 6 am and 9 am

e Channel 3 between 9 pm and midnight

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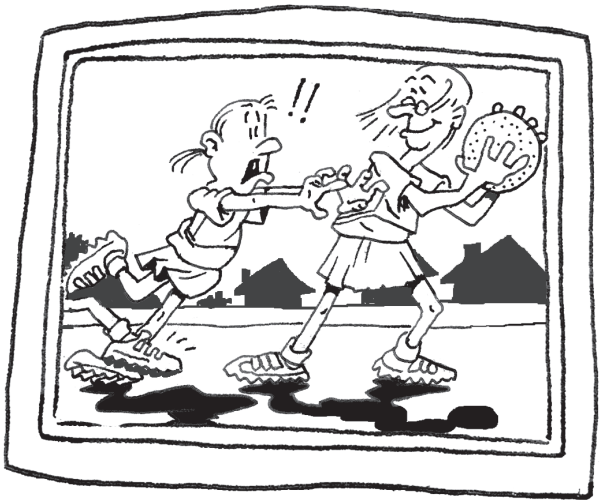
3 With TV ratings, the survey figures are extrapolated to provide a measure of how many people in total are watching different programs at different times of the day. What does extrapolate mean?

4 The city to which these survey results apply has a population of 3.5 million. In order to calculate the total estimated number of viewers in the entire city at any given time, the number of surveyed viewers is multiplied by a constant. This constant is calculated by using the following equation:

$$\text{Constant} = \frac{\text{total population}}{\text{sample size}}$$

a Calculate the constant that applies to the example used in TV ratings 1.

b Why is this number called a constant?



5 To calculate total estimated number of viewers in the entire city at any time, the following formula is used. Using your five answers to Question 2, calculate the total estimated number of viewers for each of the channels and timeslots.

$T_{actual} = C \times S$

where T = total (actual) number of viewers
C = constant
S = number of viewers from the survey

6 **a** What was the total estimated number of viewers in the entire city for the major event televised on Saturday afternoon?

b How many more viewers were estimated to be watching this event than the next most popular program during that timeslot?

7 According to these survey results, how many more people (estimated viewers in the entire city) were watching channel 5 than were watching channel 3 between 6 pm and 9 pm?

8 In your workbook, construct a line graph for each of the commercial channels (3, 4, 5 & 6) showing the percentage of viewing audience that each one attracted for the different timeslots. Use a different colour or line style for each of the four channels.

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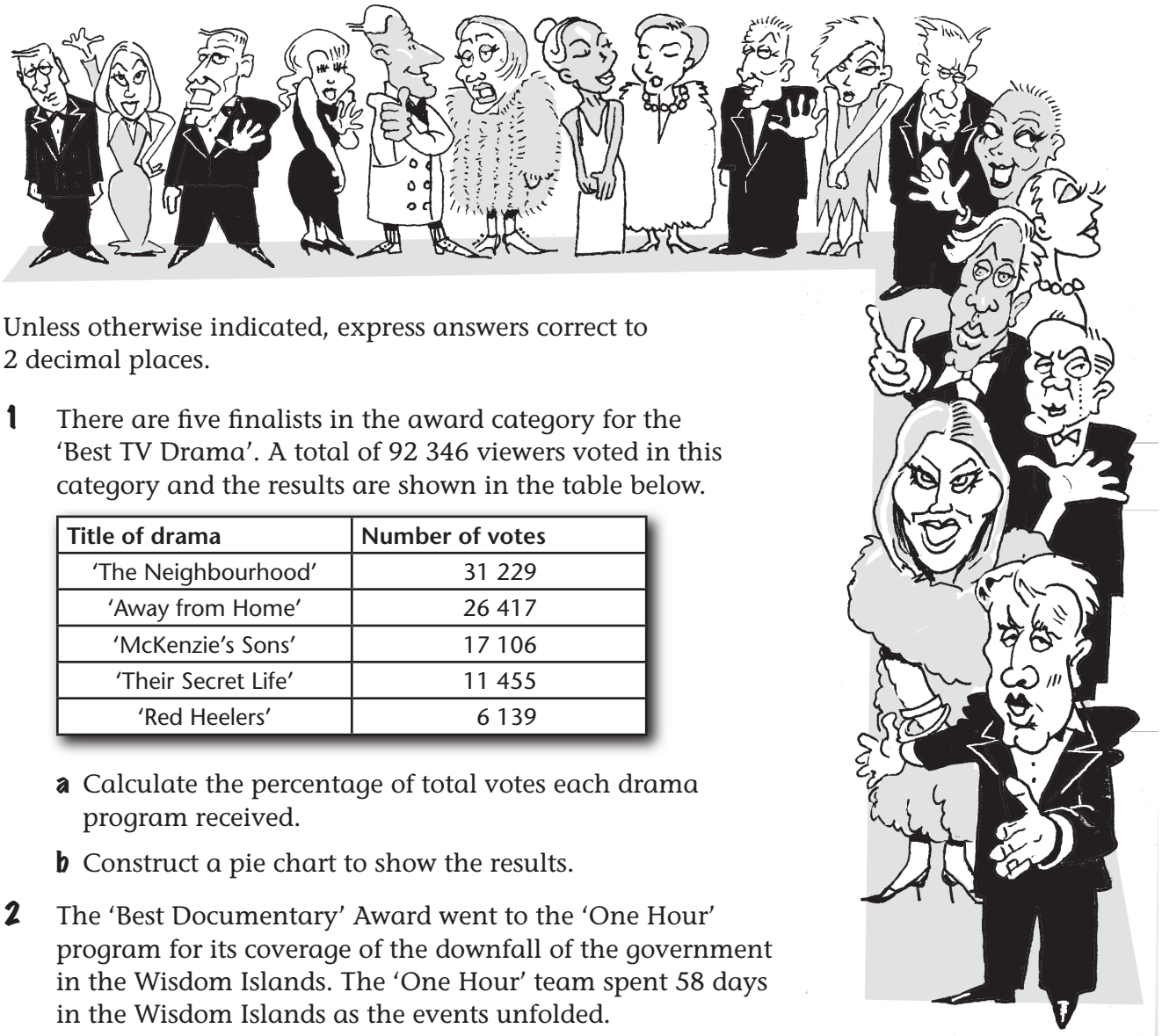
'The Golden Glitz Awards'

Television's once-a-year celebrity bash where the industry recognises its best – viewer voting, the costs of staying at the top in current affairs, speech timing, logical thinking and catering arrangements.

Television's night of nights is the once-a-year Glitz Awards, where the industry shows off its best.



Unless otherwise indicated, express answers correct to 2 decimal places.

1 There are five finalists in the award category for the 'Best TV Drama'. A total of 92 346 viewers voted in this category and the results are shown in the table below.

Title of drama	Number of votes
'The Neighbourhood'	31 229
'Away from Home'	26 417
'McKenzie's Sons'	17 106
'Their Secret Life'	11 455
'Red Heelers'	6 139

a Calculate the percentage of total votes each drama program received.

b Construct a pie chart to show the results.

2 The 'Best Documentary' Award went to the 'One Hour' program for its coverage of the downfall of the government in the Wisdom Islands. The 'One Hour' team spent 58 days in the Wisdom Islands as the events unfolded.

The team consisted of three camera operators, two sound technicians, a producer, assistant producer, two presenters, two equipment technicians and 12 other staff. In dispatching the team to the Wisdom Islands, the cost breakdown was as follows:

- wages paid to camera operators and sound technicians = $250/day
- wages paid to producer = $300/day
- wages paid to assistant producer = $275/day
- wages paid to presenters = $400/day
- wages paid to equipment technicians = $280/day
- wages paid to 12 other staff = $200/day
- return flights for each team member = $850
- accommodation, meals for each team member = $85/day

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- transport within the Wisdom Islands = $2600 (total for 58 days)
- cost of equipment and technical repairs over 58 days = $9320
- other expenses = $9440

a What was the total cost of having the 'One Hour' team stay in the Wisdom Islands and cover the story?

b If a budget of $425 000 was set aside for sending the team to the Wisdom Islands, was the actual total cost under or over budget? By how much?

3 All the attendees at the Awards were instructed that if they won an award, the maximum time they could spend on a thank-you speech was 45 seconds.

a If there were a total of 48 awards made and all speech-makers stuck to their limit, how many minutes would the speeches take up over the evening?

b One-eighth of the award winners went over the allotted speech time by an average of 18.5 seconds each. However, five award winners declined to make any speech, and another 14 of them went under the allotted speech time by an average of 11 seconds each. What was the actual amount of total speech time on the night? (Answer in minutes and seconds.)

4 The top individual award for the night, the 'Golden Glitz Award', was decided between six finalists. The runner-up received the 'Silver Glitz Award'. The finalists were:

- Lee Tan (Lead male actor in drama series 'Away from Home')
- Ralph Smithers (Host of 'Ralph Live')
- Denzel Elliot (Lead male actor in drama series 'The Neighbourhood')
- Jessica Flaherty (Presenter 'Time Off')
- Rowena Minicelli (Lead female actor in drama series 'Away from Home')
- Colleen O'Brien (Co-host of 'The Tomorrow Show')

  - Lee Tan was the second highest placed male in the 'Golden Glitz'.
  - One of the top two awards went to a lead actor in a drama series, but not the other top award.
  - Two females finished lower in the final voting than the lowest placed male.
  - Ralph Smithers won one of the top two awards last year, but finished one place lower in the voting this year.
  - It was Jessica Flaherty's first time in the final six, but the person who came fifth had also come fifth last year.

Use this information to rank the finalists in order from 1 to 6 and identify the winners of the 'Golden Glitz Award' and 'Silver Glitz Award'.

5 A total of 920 invitations were sent to TV industry people to attend the 'Glitz Awards'. How many attended the night if:

- 5% of those invited were unable to attend.
- All but 15 people who attended brought a partner with them.

6 The Awards were held at the Tiara Casino's Major Functions Room. There were 16 settings at every table. How many tables were needed to accommodate the guests?

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Answers

1 'Little Sister House' 1

1 **a** 21.8 years = 21 years 10 months
b 23.2 years = 23 years 2 months
c 22.5 years = 22 years 6 months
d 0.25 **e** 3:17 **f** 12:8 = 3:2
2 Four housemates
3 Isabel, Marlon, Miffi, Reece, Anton, Sandra
4 **a** Reece, Sandra **b** Isabel, Marlon, Miffi, Anton
c 20% **d** Wendy and Reece
e Reece 43 890; Kelvin 28 215; Wendy 53 295; Sandra 31 350 **f** 15 675 votes
5 **a** 0.60 **b** 5 : 7
6 4 new housemates
7 1 male, 3 females

2 'Little Sister House' 2

1 **a** Area living room 1: 14 m × 8.5 m = 119 m^2; Area living room 2: 8.5 m × 11 m = 93.50 m^2; Total area living rooms: 119 m^2 + 93.50 m^2 = 212.50 m^2
b 119 m^2 / 20 = 5.95 m^2
2 Neither, they are the same area, 34 m^2.
3 16.38 m **4** 329 m^2 **5** 90.57 = 91 cameras
6 150 m
7 **a** 16 seconds per lap **b** 87.50 m **c** 0.78 m/s
8 **a** 127.50 m^3 b 127 500 L
9 **a** NE **b** S **c** W **d** N **e** NW

3 Game shows

1 Jasmine: $30, $75, $40, $30, $40, $75. Huan: $40, $15, $40, $20, $25, $50. Adam: $20, $15, $80, $10, $35, $70. Rank: Jasmine 1, Huan 3, Adam 2.
2 Home entertainment system $9000; Car $35 000; Lounge room furniture $4500; Mountain bike $1750; Luggage $2500; Holiday $14 000; Gym equipment $17 500
3 **a** 6 in 24 or 1 in 4 **b** 3 in 24 or 1 in 8
c 2 in 24 or 1 in 12 **d** 8 in 24 or 1 in 3
e Dud prize, 8 in 24 or 1 in 3 against 6 in 24 or 1 in 4
f Four times, 4 in 24 or 1 in 6 against 1 in 24
4 **a** $7200 **b** $50 000 **c** $185 000 **d** $75 000

4 'The Best Loser' 1

1 Because people are built differently, e.g. males of a particular height usually weigh more than females of the same height. There are also many other genetic and other differences that account for varying body weights.
2 Teacher to check graphs.
3 Stefan 45 kg; Frank 42 kg; Mohammed 34 kg; Susan 32 kg; Heidi 23 kg
4 **a & b** Susan 37.6%; Stefan 35.1%; Frank 31.6%; Mohammed 29.6%; Heidi 25.3%
5 Stefan 7.5 kg/month; Frank 7.0 kg/month; Mohammed 5.7 kg/month; Susan 5.3 kg/month; Heidi 3.8 kg/month
6 **a** Divide the figure in centimetres by 100, or move the decimal point two places to the left.
b BMI at initial weight: Susan 36.3; Heidi 32.6; Stefan 39.9; Mohammed 35.1; Frank 35.3
c BMI at target weight: Susan 22.6; Heidi 24.4; Stefan 25.9; Mohammed 24.7; Frank 24.2
7 **a** All contestants fitted into the obese category.
b No, Stefan would be just outside it in the overweight category.
8 Teacher to check graphs.

5 'The Best Loser' 2

1 Susan 4.5 kg, Heidi 4 kg, Stefan 12 kg, Mohammed 5 kg, Frank 16 kg
2 Frank, Stefan, Mohammed, Susan, Heidi
3 Heidi: 1 hour of cycling 91 × 35 × 1 = 3185, 4.5 hours of sitting 91 × 4 × 3.5 = 1638, 1.5 hours of slow walking 91 × 17 × 1 = 2320.5, 4 hours of standing 91 × 6 × 4 = 2184, TOTAL 9327.5kJ so weight is lost.
Stefan: 4 hours of golf 128 × 19.5 × 4 = 9985, 1 hour of tennis 128 × 26 × 1 = 3328, 5 hours of sitting 128 × 4 × 5 = 2560, 3.5 hours of standing 128 × 6 × 3.5 = 2688. TOTAL 18561kJ so weight is lost.
Frank: 0.5 hours of jogging 133 × 37 × 0.5 = 2560.5, 0.25 hours of brisk walking 133 × 20 × 0.25 = 665, 6.5 hours of sitting 133 × 4 × 6.5 = 3568, 2 hours of standing 133 × 6 × 2 = 1596, TOTAL 8389.5kJ so weight is gained.
4 Heidi – 87.5 kJ more energy expended than taken in; Stefan – 7811 kJ more energy expended than taken in; Frank – 4110.5 kJ more energy taken in than expended.
5 Teacher to check graphs.

6 Advertising

1 **a** Advertising rates vary because cities and regions have different populations – the more viewers an advertisement reaches, the higher the price the advertiser pays.
b There are different numbers of viewers at different times of the day e.g. during a weekday fewer people are watching television because they are at work or school. Again, the more viewers an advertisement reaches, the higher the price the advertiser pays.
2 Melbourne $633.33; Sydney $716.67; Brisbane $473.33; Adelaide $173.33; Perth $223.33
3 Three times
4 15 s: $8520, $284.
45 s: $19 880, $662.67.
60 s: $22 720, $757.33.
5 $6700 + $6030 + $4221 = $16 951 total cost
6 **a** Yes **b** $6372 – $5200 = $1172
7 **a** Ad running time = 1170 s = 19.50 min; Movie running time = 108 min; Total time = 127.50 min, meaning 7.50 min need to be cut from the movie.
b This is 6.94% of the total movie time.

7 'National Icon'

1 **a** NSW = 536; Vic. = 554; Qld = 396; WA = 389; SA = 312; Tas. = 108; ACT = 63; NT = 42
b 2400 **c** 1273 **d** 1127
2 **a** 1273 : 1127 = 1 : 1.13 **b** ACT **c** NSW
3 0.083 **4** 92.7% **5** 1 in 200
6 The ratio of females to males on the audition list of 200 is greater than the ratio of females to males among the initial applicants. Audition list ratio females to males = 119 : 81 = 1 : 0.68. Initial ratio females to males = 1273 : 1127 = 1 : 1.13.
7 0.50%
8 **a** d = P – p; p = P – d **b** P = S/c; c = S/P
9

Grace	12.5	16.0	28.5	1.8	51.3
Felix	14.0	18.5	32.5	2.6	84.5
Bryant	16.5	14.5	31.0	2.2	68.2
Stacey	12.0	16.5	28.5	2.3	65.6
Giovanni	16.0	17.5	33.5	2.7	90.5
Rachel	14.5	15.0	29.5	1.7	50.2
Colin	14.5	10.0	24.5	2.6	63.7
Ebony	11.5	11.5	23.0	2.5	57.5
Fatima	16.0	13.5	29.5	2.4	70.8
Matthew	18.5	19.0	37.5	1.9	71.3
Lucien	12.5	17.5	30.0	1.9	57.0
Holly	11.5	9.5	21.0	2.8	58.8

10 From highest to lowest: Giovanni; Felix; Matthew; Fatima; Bryant; Stacey; Colin; Holly; Ebony; Lucien; Grace; Rachel

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Answers (cont'd)

8 'Out in the Wild'

1 **a** 65 **b** 32 **c** 101 **d** 3.2
2 **a** 7 : 5 or 1.4 : 1 **b** 0.93 **c** 13 cm **d** 58.5%
3 **a** 15 months **b** Weight = 4.5%; height = 35%
4 **a** 54.6 km **b** 2847 km **c** 1.62 km

9 'Backyard Bonanza'

1 1289.61 m^2 **2** 96.10 m^2 **3** 1205 pavers (1204.3)
4 9.6 hours = 9 hours 36 minutes
5 A pergola – rectangle; B permanent BBQ (with paving) – square BBQ, circle paving stones; C gazebo – octagon; D shade sail – triangle; E children's sandpit – rectangle; F fish pond – oval; G bordered native grass plantation – trapezium
6 $50.76
7 **a** 37.20 m^2 **b** 1.9 m^3 **c** $77.75

10 Here is the News

1 **a** 5330 burglaries **b** 8.2%
2 $3596
3 **a** $5 352 941 per kilometre
b $272 999 999 ($273 million)
4 **a** 1640 kg **b** 1174 kg **c** 368 cm
d 4.01 tonnes **e** 2.37 tonnes heavier
5 **a** 0.31
b Year 1 = $192 000; Year 2 = $208 000; Year 3 = $352 000; Year 4 = $352 000; Year 5 = $496 000
6 **a** 104 = 10 000 times
b Approximately 313 or 29 791 times more energy.
c This means that one such earthquake occurs on average every 3.04 days in a year.
7 **a** $2 735 420 **b** $204 462 **c** $1522 **d** $115
e $3 419 275 **f** $683 855
8 Minimum temperature = 55.4°F; Maximum temperature = 75.2°F

11 'What's it Worth?'

1 The pair of vases is worth $374 more than the owners paid for them.
2 $23 559
3 **a** 1896 **b** $883.50
c Because a set of six is potentially more useful to its owner than individual bowls. Collectors are generally willing to pay more for collectable items in a set than they are for individual pieces.
4 **a** 472 **b** 10 156 containers
5 **a** The minimum price that the owner of an auctioned item is willing to sell for.
b $1915 **c** Yes it sold.
d It exceeded its reserve price by $115.
e It was 6.4% above the reserve price.
6 **a** 4.15 m^2 **b** 5.99 m^2 **c** 0.80 m or 80.00 cm

12 'Funny Videos Shot at Home'

1 **a** 32° **b** 6.25 m/s **c** 36.1 m
2 **a** 1.84 m **b** 4.49 times her own height
3 **a** 79.13 m^2 **b** 37.69 m
4 Rear guy rope at 49° – within recommended range; Front guy rope at 29° – not within recommended range.
5 **a** 18.11 m^2 **b** $93.28

13 'Alan Forsyth Live'

1 96° **2** 104°
3 **a** A = 33°, B = 33°, C = 147°, D = 147°, E = 33°, F = 33°
b Angles E and F **c** Angles B and C
4 **a** Trapezium **b** A = 78°, B = 288°
5 72 females : 60 males
6 **a** 33 questions **b** 11 questions per guest
c First guest = 1.83, Second guest = 2.33, Third guest = 1.33
7 **a** 15 min **b** 25% **c** 3 min
d 6 advertisements

14 'Cops on the Beat'

1 **a** $14 231
b Min. = $16 500 – 40% = $9900; Max = $17 000 – 30% = $11 900
c 11.1%
2 **a** $258 230 **b** $344.60 **c** 17.3%
d $215 460 **e** $117 460
3 **a** 532 **b** 157 **c** 136
d Various points of view could be put forward, but the evidence suggests that a small number of people were involved in collecting most of the undersized fish, and that they had exceeded catch limits.
4 **a** 1% and 0.5% respectively
b He was treated fairly, as there is evidence he attempted to measure his catch, and the two that are technically undersized fall short by only a small amount.
c To get himself a new and more accurate device with which to measure his catch
5 Teacher to check column graphs and pie charts.

15 'Time Off!'

1 **a** 3 h 25 min **b** 2562 km
c 936.3 km/h (if calculated with a fraction for time, i.e. 22/3 h, then answer is 937.5 km/h)
d 82.7 km/h **e** 749.8 km/h
2 **a** 254.47 m^2 **b** 246.97 m^2 **c** 56.55 m
d 381.75 m^3 = 381 750 L
3 $2340 **4** 41%
5 **a** 320 L **b** $560 **c** 2783 km
6 **a & b** Teacher to check.
c 1 : 1 000 000
7 35°54'N 14°31'E
8 4.30 am on Thursday
9 $3159
10 20 334.6 = 20 335 beds

16 'My Reality Restaurant'

1 Alex & Hayley: $1430, $2200, $22.50, 290, $6525, 6525 – 3630 = $2895, $960, $1935.
Rhonda & Issa: $1890, $1950, $18.50, 345, $6382.50, 6382.50 – 3840 = $2542.50, $580, $1962.50.
Carlos & Mae: $2450, $2975, $24.70, 400, $9880, 9880 – 5425 = $4425, $1125, $3300.
Inez & Victor: $1680, $2100, $17.95, 325, $5833.75 5833.75 – 3780 = 2053.75, $620, $1433.75.
2 **a** Carlos & Mae; Rhonda & Issa; Alex & Hayley; Inez & Victor
b No, Rhonda & Issa had a lower gross profit figure, but a higher net profit.
c

Cost factor	Effect on net profit
Increase cost of food	Decrease
Decrease cost of food	Increase
Increase cost of wages	Decrease
Decrease cost of wages	Increase
Increase av. cost of meal	Increase
Decrease av. cost of meal	Decrease
Increase projected no. of meals	Increase
Decrease projected no. of meals	Decrease
Increase other running costs	Decrease
Decrease other running costs	Increase

3 Alex & Hayley 25 meals short; Rhonda & Issa 48 meals over; Carlos & Mae 32 meals short; Victor & Inez 52 meals over.
4 Alex & Hayley –8.6%; Rhonda & Issa +13.9%; Carlos & Mae –8%; Victor & Inez +16%. Carlos & Mae were the closest in their projection.

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5 Alex & Hayley: $1430, $2200, $22.50, 265, $5962.50, $2332.50, $960, 1372.50.
Rhonda & Issa: $1890, $1950, $18.50, 393, $7270.50, $3430.50, $580, $2850.50.
Carlos & Mae: $2450, $2975, $24.70, 368, $9089.60, $3664.60, $1125, $2539.60
Inez & Victor:$1680, $2100, $17.95, 377, $6767.15, $2987.15, $620, 2367.15

6 Teacher to check graphs.

17 'Wonderful World of Sports'

1 a Johnson (Aus) 206; Gullan (India) 207; Robinson (USA) 207; Villani (Italy) 209; Jorgensen (Sweden) 210; Katazumi (Japan) 210; Anderson (Aus) 211; Li (China) 211; Broderick (S. Africa) 211; Hansen (USA) 212

b 67 **c** No playoff was needed to decide the winner.

2

Player	Kicks	H'balls	Possessions
Stewart (Cheetahs)	20	12	32
Henderson (Falcons)	12	18	30
Iskra (Falcons)	21	7	28
Samuels (Falcons)	13	13	26
D'Amor (Cheetahs)	12	12	24

3 a 8 laps between them **b** 2.95 s **c** 1.37.69

4 99.24 metres

18 Logging and editing

1 a 2 hours, 6 minutes, 17.08 seconds

b 28 minutes **c** 0.04 or 4/100 of a second

d 17 hours, 19 minutes, 45.12 seconds

2 48.44 seconds

3 a Game1

Tape no.	Start	Finish	Total
1	00:04:12:50	00:04:19:40	6.90 s
	00:11:45:09	00:12:18:75	33.66 s
	00:33:57:00	00:34:18:19	21.19 s
	01:09:55:66	01:10:09:81	14.15 s
TOTAL HIGHLIGHTS GAME 1 =			75.90 s
			=1 min 15.90 s
Game 2			
1	00:23:36:47	00:23:52:06	15.59 s
2	00:07:42:45	00:08:00:02	17.57 s
	00:48:55:98	00:49:11:31	15.33 s
	01:17:58:14	01:18:08:49	10.35 s
TOTAL HIGHLIGHTS GAME 2 (min & s) =			58.84 s

b Game 1 **c** 17.06 seconds more

4 a 5 tapes **b** 5.9%

5 6.33 seconds

6 a 22 min 15.33 s **b** 17.6%

c 1 h 17 min 2.56 s of footage is still available.

19 TV ratings 1

1 a Population – the entire group of people to whom a survey is relevant.

b Sample – the group analysed, which is chosen from the population.

c Sample size – the number of people in the sample.

2 a Because some programs are different in various states, e.g. the news; the type of sport (particularly football codes) in certain states; coverage of major city and state events, e.g. the ANZAC Day parade in different capital cities, and so on.

b The vast majority of Australia's population lives in the state capital cities.

3 a The population of the city.

b The people living in the recruited households.

c 5000 viewers

d Yes, 5000 people is quite a large number of people to survey.

e 0.02% of 3 500 000 = 700

f No, it is a very small number, and those in the category of never watching any television at all are unlikely to live in a household with a television.

4 a 6 pm to 9 pm because 78% of viewers are watching then.

b Evening meal time, many more people are home rather than out doing other things.

c The rates charged to advertisers would be greater during the timeslots when more people are watching TV.

d Channel 3 **e** 1118

f The advertising rates would be at a premium, because of the high number of viewers.

20 TV ratings 2

1 a 950 viewers **b** 1750 viewers **c** 2625 viewers

d 3900 viewers **e** 1900 viewers **f** 400 viewers

2 a 158 viewers **b** 78 viewers **c** 239 viewers

d 292 viewers **e** 411 viewers

3 Extrapolate means to make an estimate or prediction about something using information that is already known. In mathematics, this means estimating a larger (or smaller) quantity or value by using a known smaller (or larger) value.

4 a 700

b It is called a constant because it is always the same number that is used as the multiplier.

5 a 110 600 **b** 54 600 **c** 167 300 **d** 204 400

e 287 700

6 a 782 775 viewers **b** 341 775

7 87 360 more viewers **8** Teacher to check graphs.

21 'The Golden Glitz Awards'

1 a total of 100%: 'The Neighbourhood' – 33.82%; 'Away from Home' – 28.61%; 'McKenzie's Sons' – 18.52%; 'Their Secret Life' – 12.40%; 'Red Heelers' – 6.65%

b Teacher to check pie charts.

2 a total cost = $478 230
total wages paid to camera operators and sound technicians = $72 500
total wages paid to producer = $17 400
total wages paid to assistant producer = $15 950
total wages paid to presenters = $46 400
total wages paid to equipment technicians = $32 480
total wages paid to 12 other staff = $139 200
total return flights for each team member = $19 550
total accommodation, meals for each team member = $113 390
transport within the Wisdom Islands = $2600
cost of equipment and technical repairs = $9320
other expenses = $9 440

b Over budget by $53 230.

3 a 36 minutes **b** 31.53 min = 31 min 32 s

4 1 Rowena Minicelli; 2 Ralph Smithers; 3 Lee Tan; 4 Denzel Elliot; 5 Colleen O'Brien; 6 Jessica Flaherty

5 1733 people attended

6 108.31 = 109 tables

Production
Commissioning editor: Jennifer Gough
Managing editor: Ruth Schultz
Cover design: Gail McManus Graphics
Editor: Brenda Hamilton
Proofreader: Judith Schultz
Illustrator: David Dickson
Layout and computer art: Final Page Editing and Design

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Have you ever been lying on the couch watching the television and thought about how much maths there is in each program? And then thought that these topics would be a great way to teach practical maths skills to your students? Well relax – we've done the thinking too and come up with a range of photocopiable worksheets that will have your students fully engaged in the chances of winning on games shows, the costs of creating a restaurant from scratch, the geometry of shooting videos and the problems of calculating sporting performances. These topics and more will motivate and challenge even the most reluctant student – and with all the answers provided, marking will be a breeze!

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- Assignments in Junior Algebra (Books 1, 2)
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